Complexity reduction in Model Predictive Control

Learning from experts with guarantees

Dinesh Krishnamoorthy

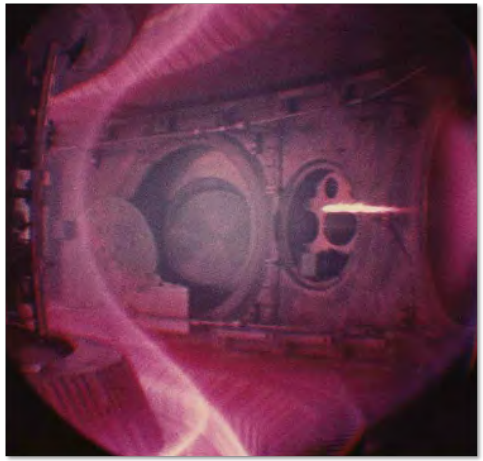
dinesh.krishnamoorthy@ntnu.no

NTNU

DCSC Seminar, 06 May 2026

Example: Fuelling Nuclear Fusion Devices

Discharge 196325 Large Helical Device, NIFS Japan.



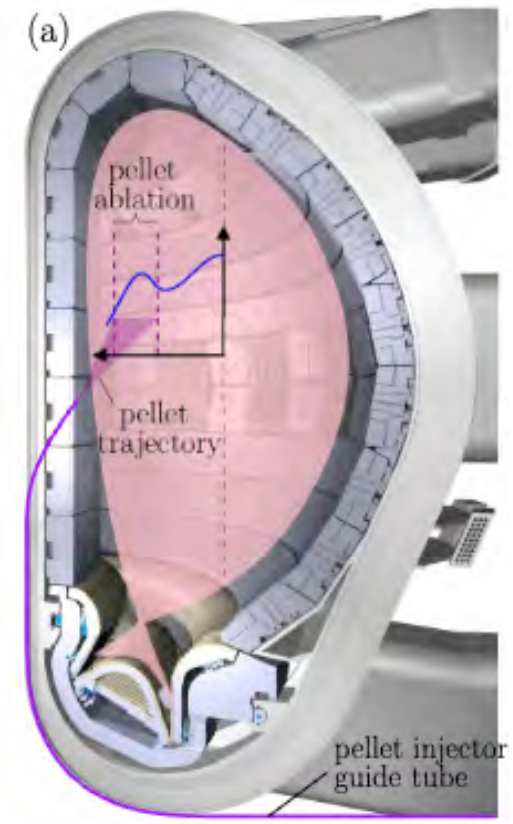


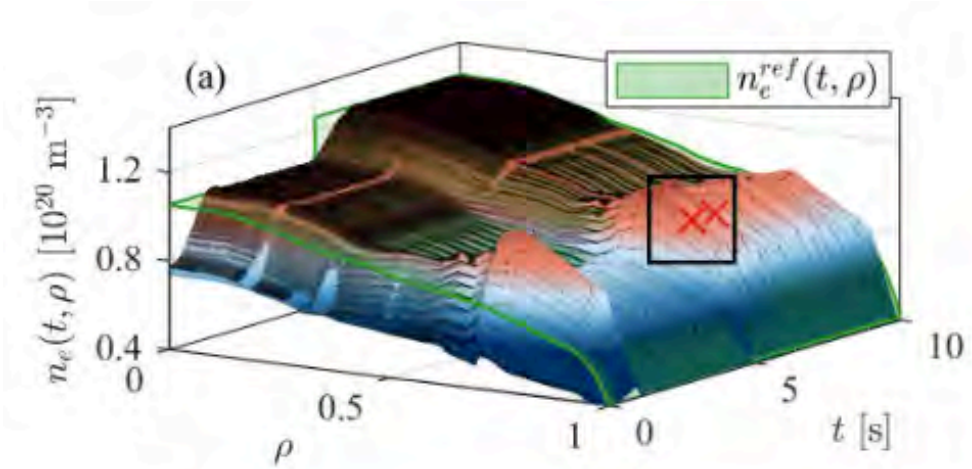
Experiments by Christopher Orrico.

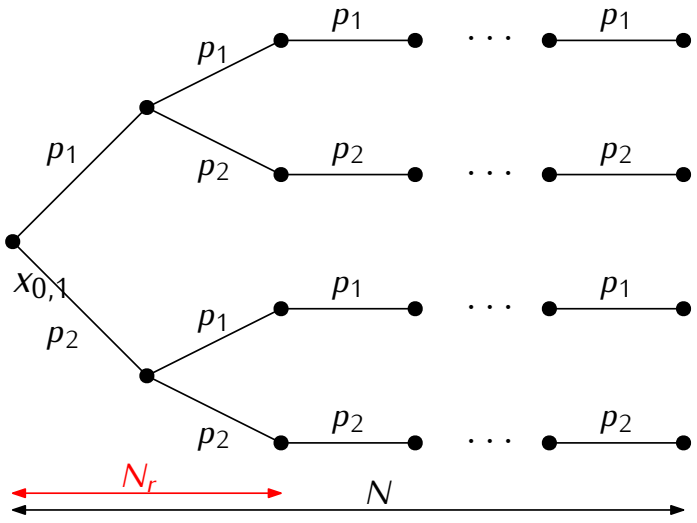


S. K. Combs and L. R. Baylor. *Fus. Sci and Tech.* (2018)

Example: Fuelling Nuclear Fusion Devices







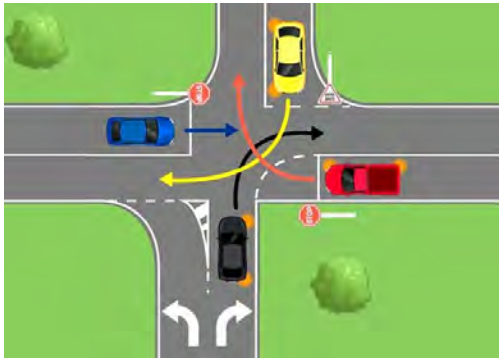
Orrico et al. *IEEE L-CSS* (2023)
Orrico et al. *IEEE CDC* (2025)
Orrico et al. *Nuclear Fusion* (2025)

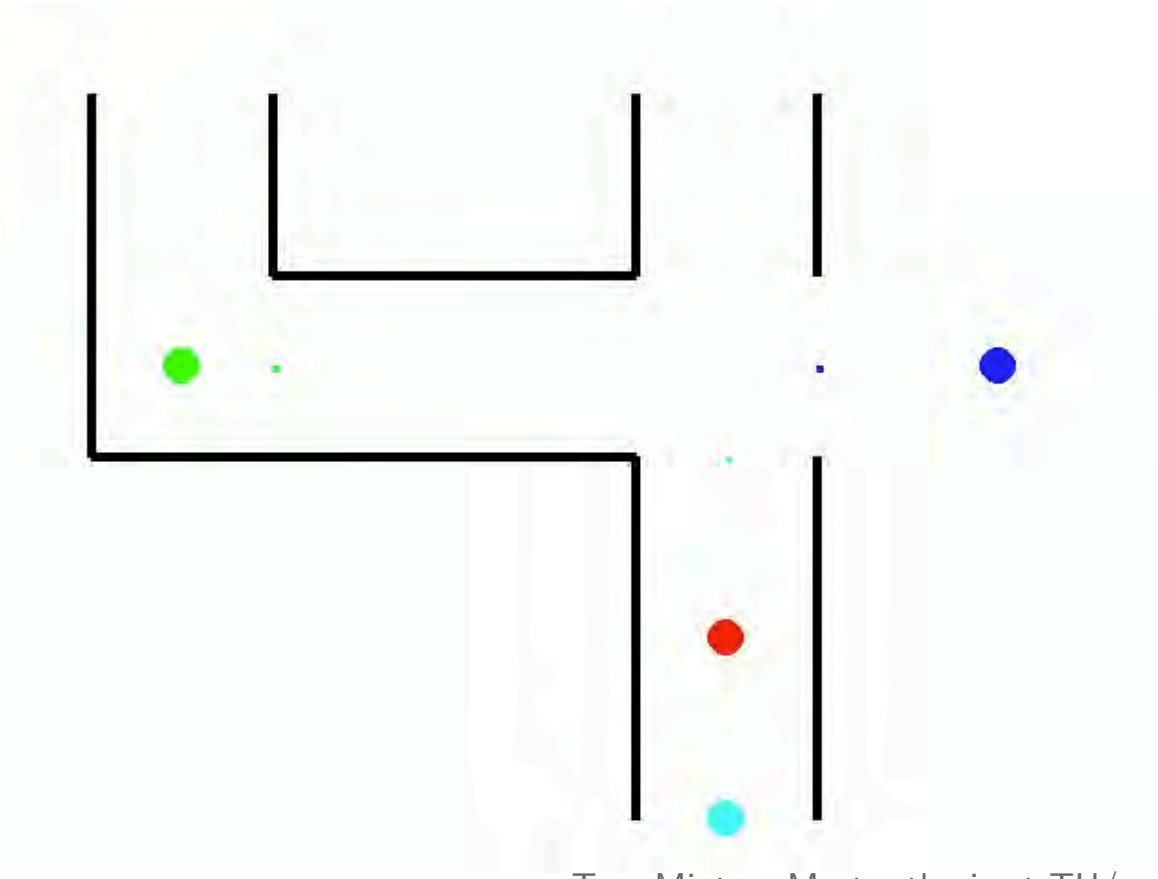
Mixed-integer MPC

Reduced complexity enable real-time mixed-integer decision-making.

Example: Traffic intersection

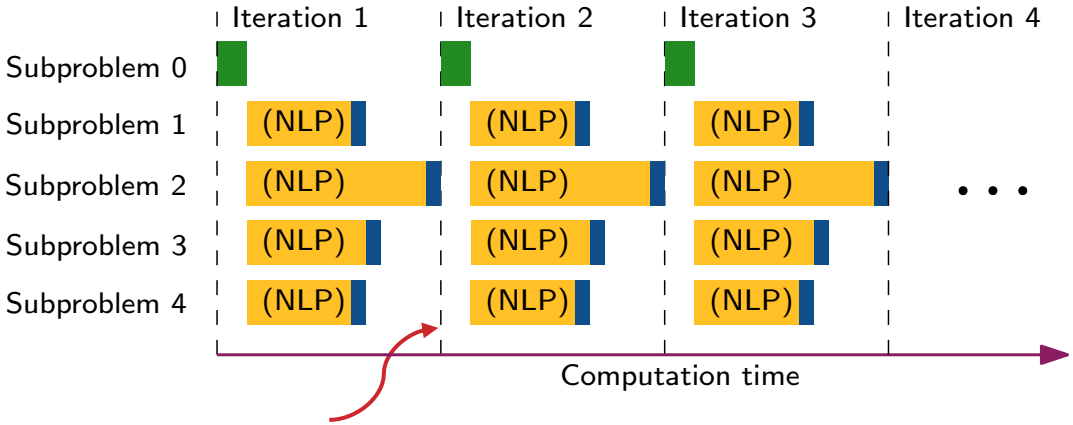






Tom Minten, Master thesis at TU/e

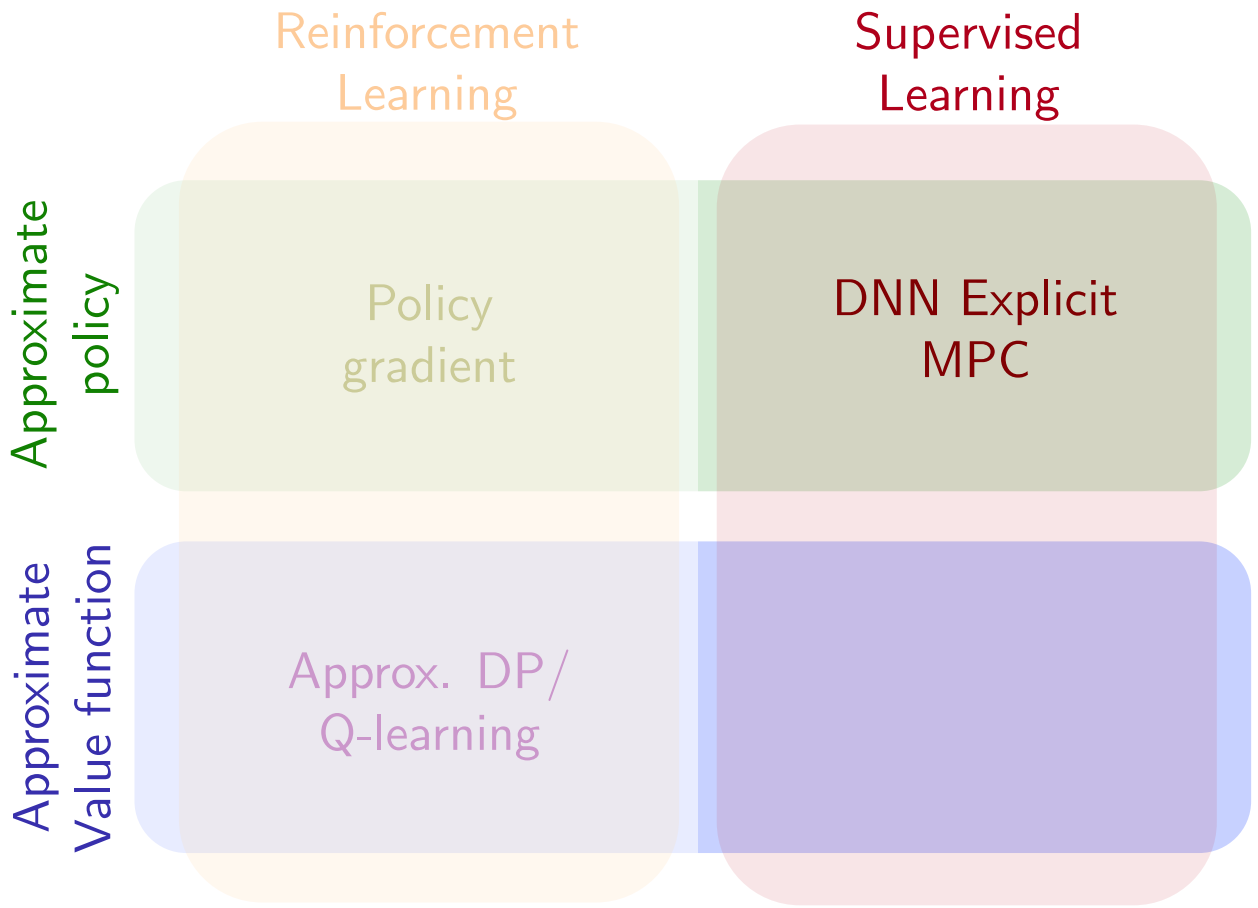
Distributed MPC



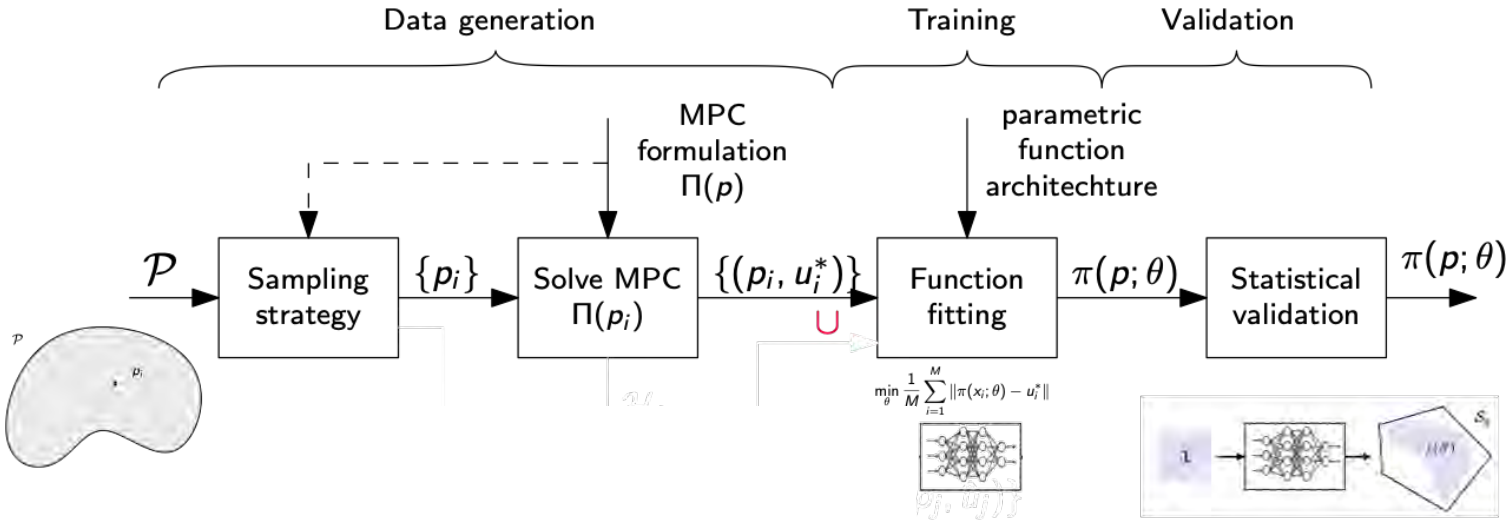
Reduced complexity of the subproblems enable real-time distributed decision-making.

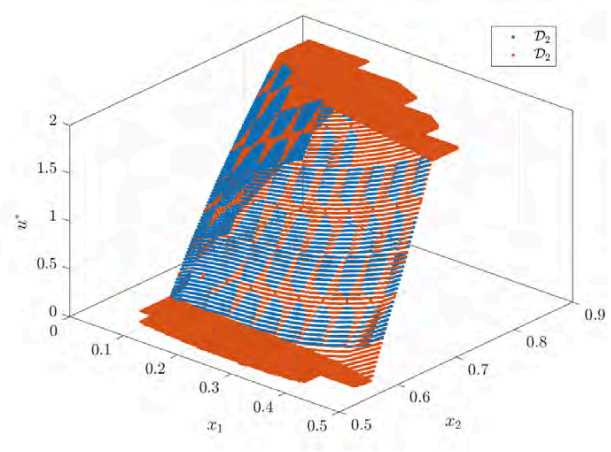
$$\mathbf{w}_i^{k+1} = \arg\min_{\mathbf{w_i}} \mathcal{L}_i(\mathbf{w_i}, p_i^k)$$
$$\text{s.t. } c_i(\mathbf{w}_i) \leq 0$$

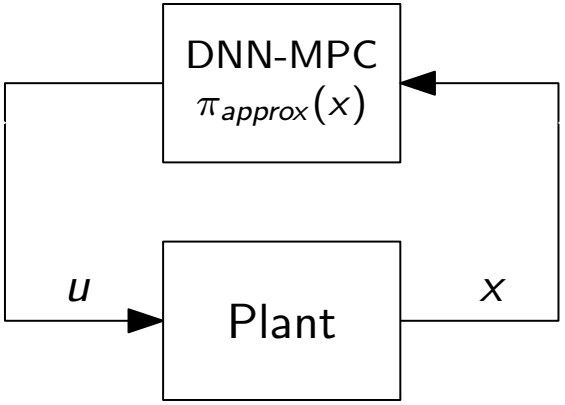
Premise: We have a good MPC problem formulation, but it is too complex for real-time implementation


Reinforcement Learning
Supervised Learning
Approximate policy
Approximate Value function
Policy gradient
DNN Explicit MPC
Approx. DP/ Q-learning

Learn optimal policy from expert data



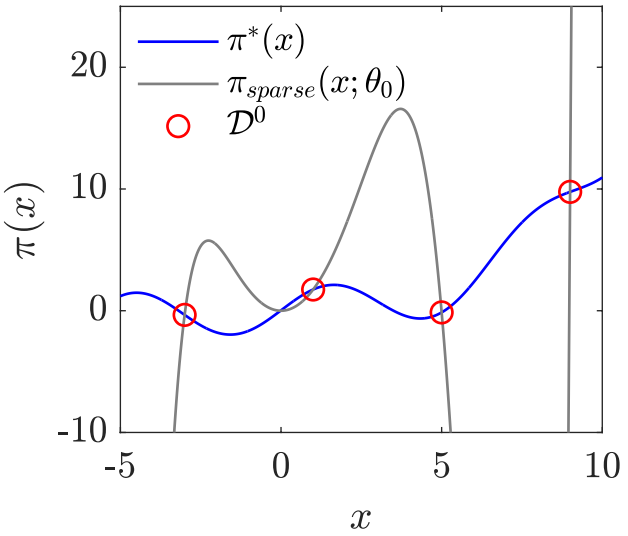


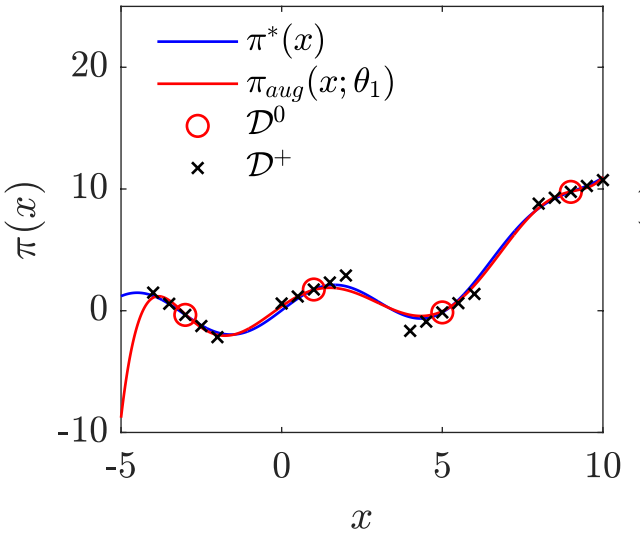


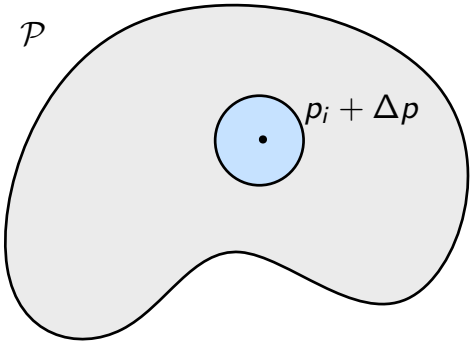
Parisini & Zoppoli (1995) *Automatica*, Chakrabarty et al. (2017) *IEEE TAC*, Karg & Lucia (2018), *IFAC NMPC*, Chen et al. (2018) *ACC* Chen et al. (2022) *Automatica*, Paulson & Mesbah (2020) *IEEE L-CSS*, Hertneck et al. (2018) *IEEE L-CSS*, Drgona et al. (2018) *Applied Energy*, Ceskö et al. (2015) *IEEE-TCST*, Zhang et al. (2019) *ACC*, Spielberg et al. (2019) *AIChE J.*

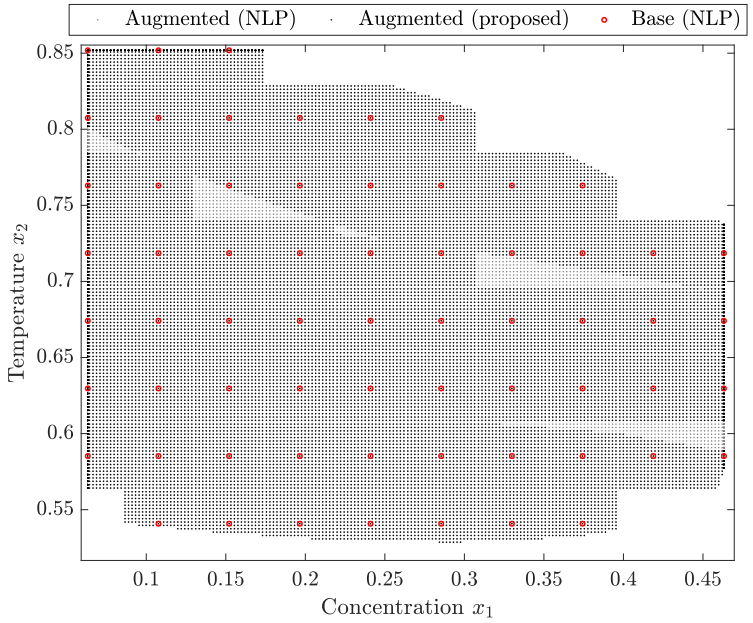
Sensitivity-based Data Augmentation











Krishnamoorthy, *IEEE Trans. Aut. Control.* (2021)

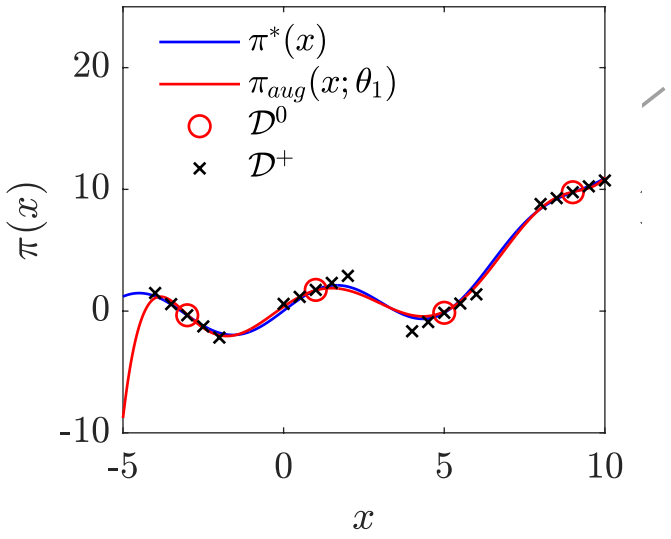
Sensitivity-based Data Augmentation

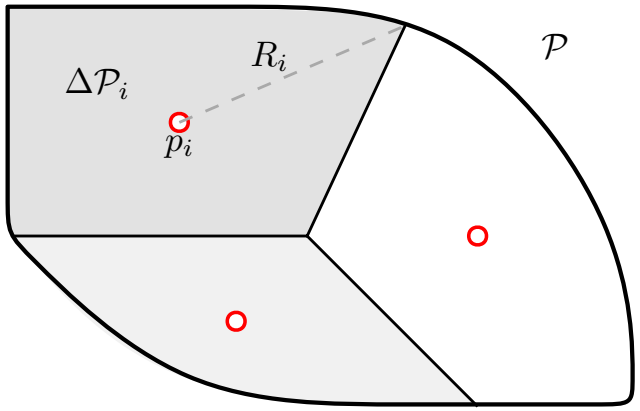
Error Bound due to augmenting with inexact samples

Assuming that the parametric function $\pi(x, \theta)$ is sufficiently richly parameterized, we have

$$\|\pi(x, \theta_1) - \pi^*(x)\| \leq \max\left(\left\{L_{x_i} R_i^2\right\}_{i=1}^{N}\right)$$

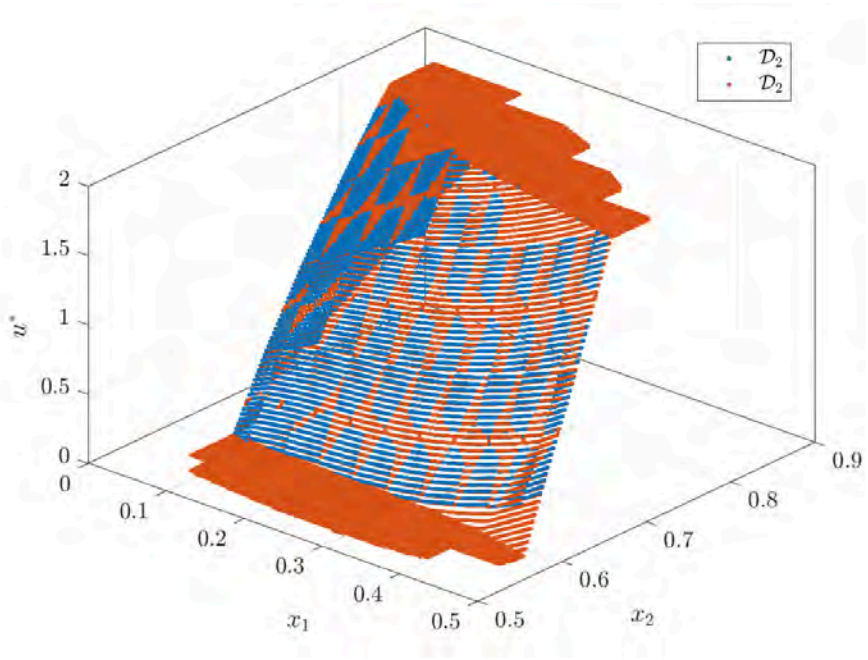
in probability as $M \to \infty$ if θ_1 is a consistent estimator.

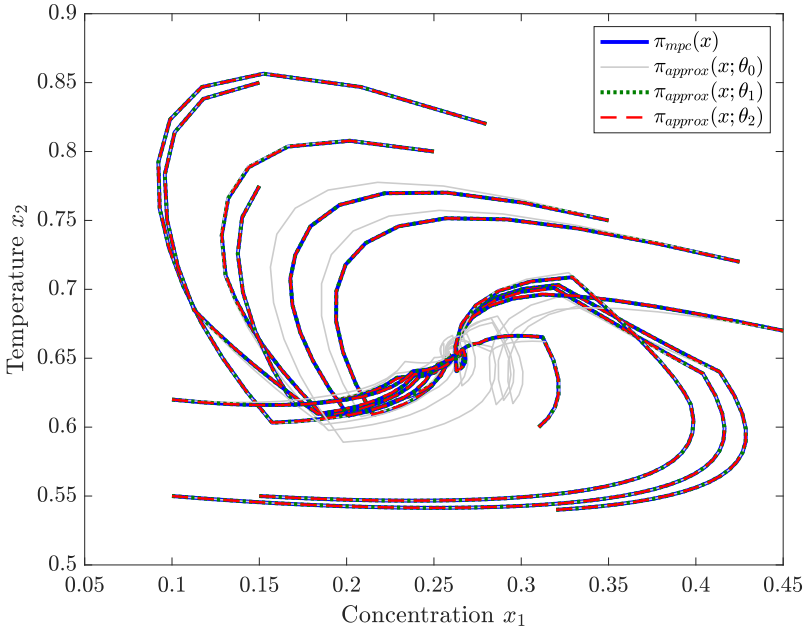




Krishnamoorthy, *IEEE Trans. Aut. Control.* (2021)

Example: Benchmark CSTR





Krishnamoorthy, *IEEE Trans. Aut. Control.* (2021)

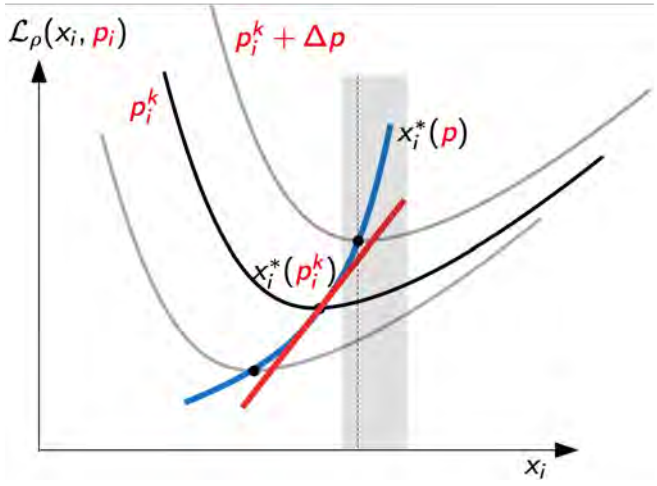
Sensitivity-based Data Augmentation

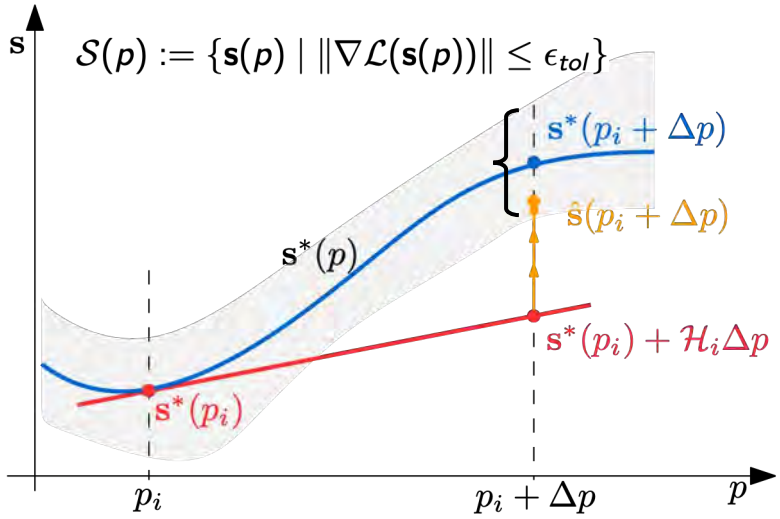
Error Bound due to augmenting with inexact samples

Assuming that the parametric function $\pi(x, \theta)$ is sufficiently richly parameterized, we have

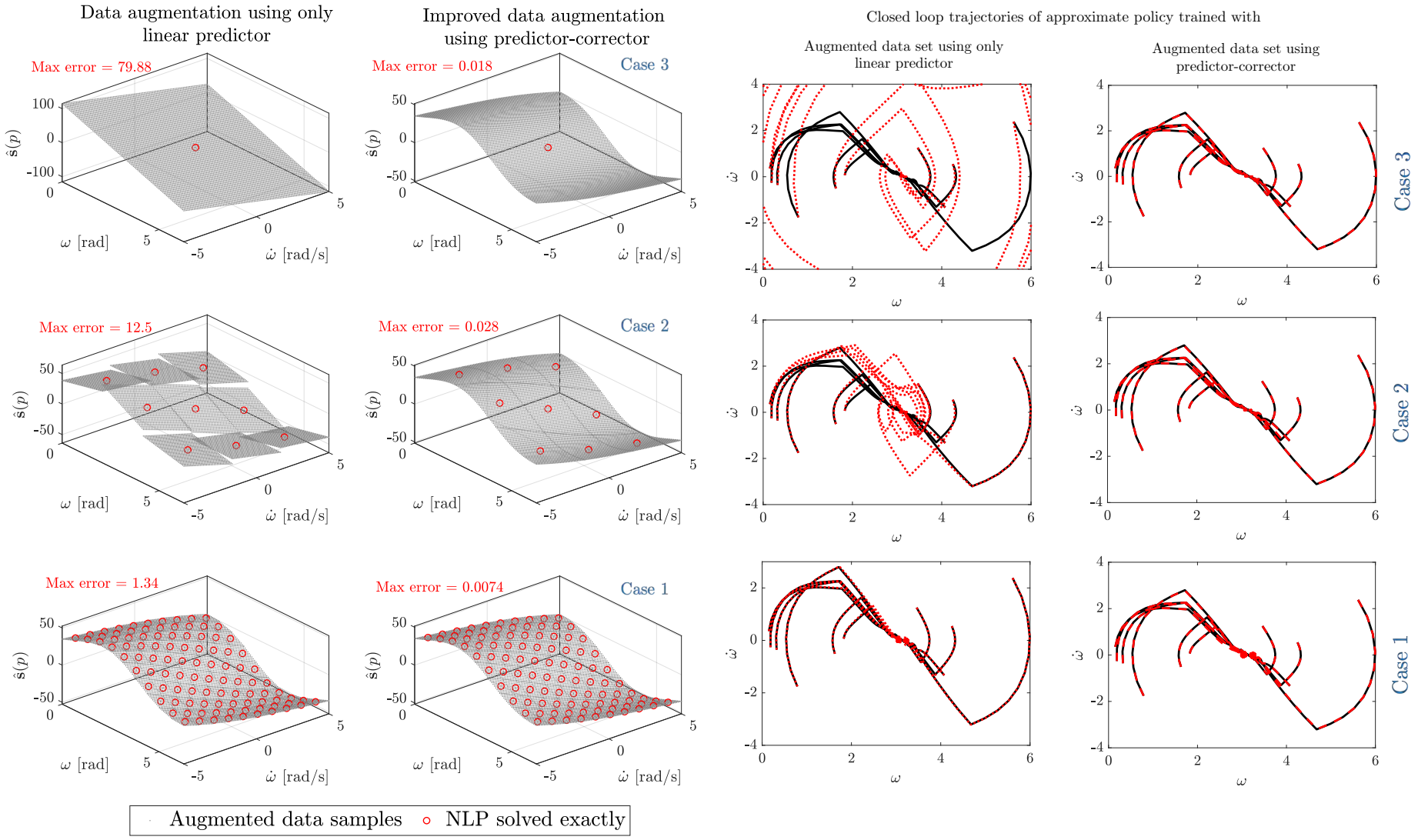
$$\|\pi(p; \hat{\theta}) - \pi^*(p)\| \leq \sup_{\mathbf{s}(p) \in \mathcal{S}(p)} \|\mathbf{s}(p) - \mathbf{s}^*(p)\|$$

in probability as $M \to \infty$ if θ_1 is a consistent estimator.

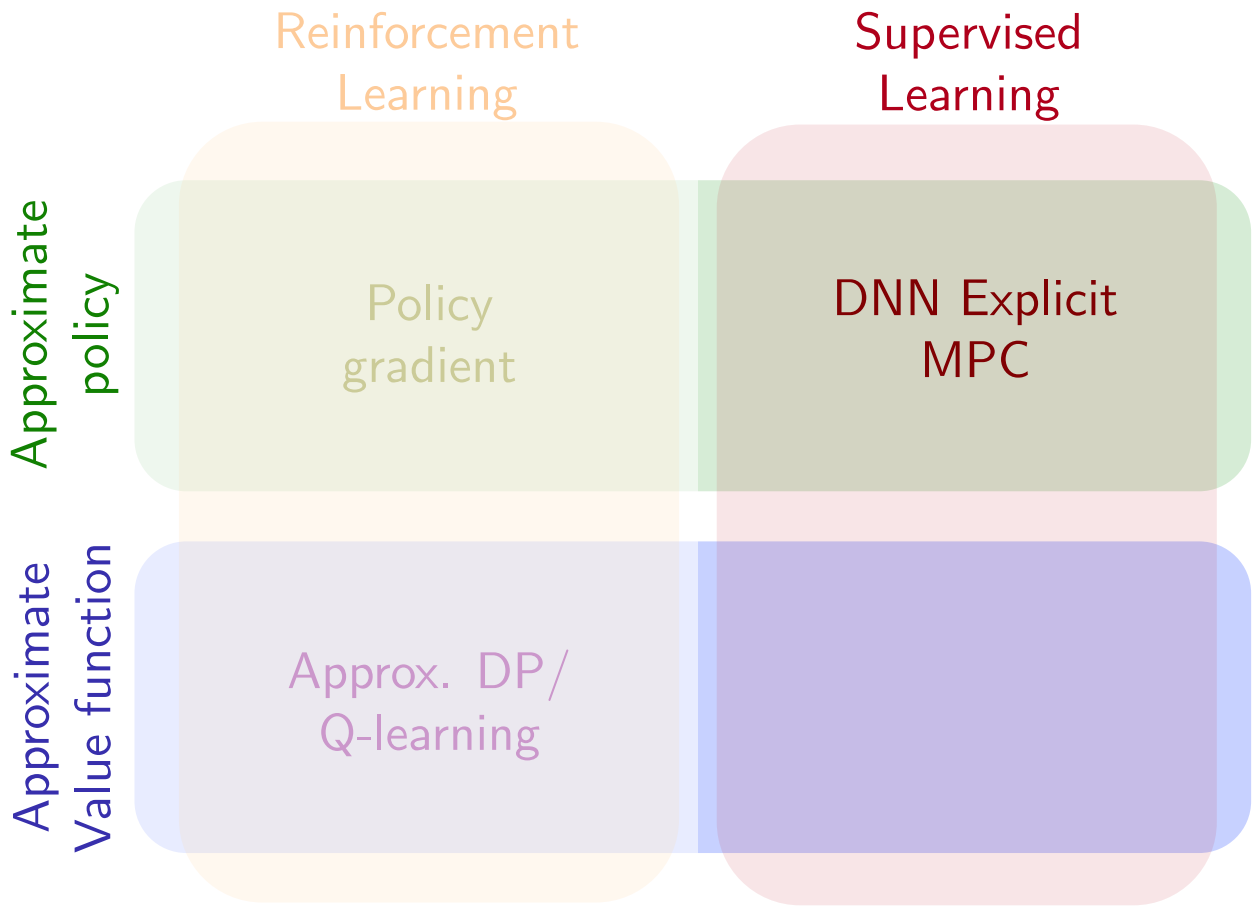




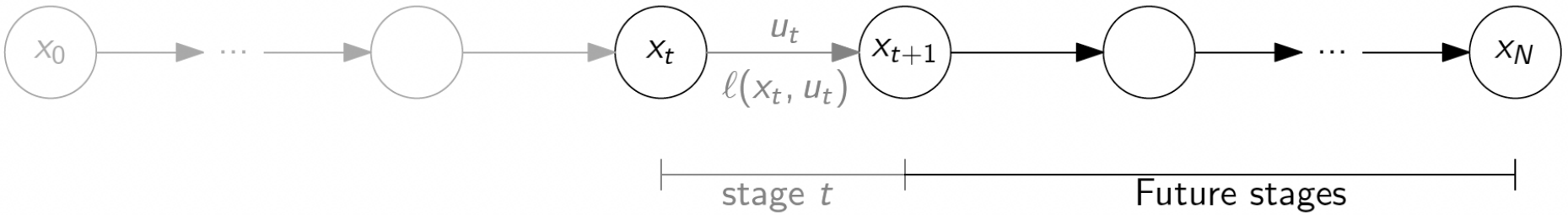
Sensitivity-based Data Augmentation



Krishnamoorthy, *IEEE L-CSS* 2024


Reinforcement Learning
Supervised Learning
Approximate policy
Approximate Value function
Policy gradient
DNN Explicit MPC
Approx. DP/ Q-learning

Sequential decision making

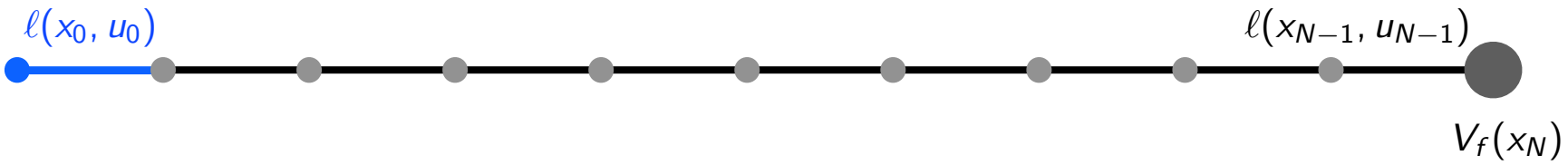


$$\pi^*(x) = \arg\min_{u \in \mathcal{U}} \ \ell(x, u) + J^*(f(x, u))$$

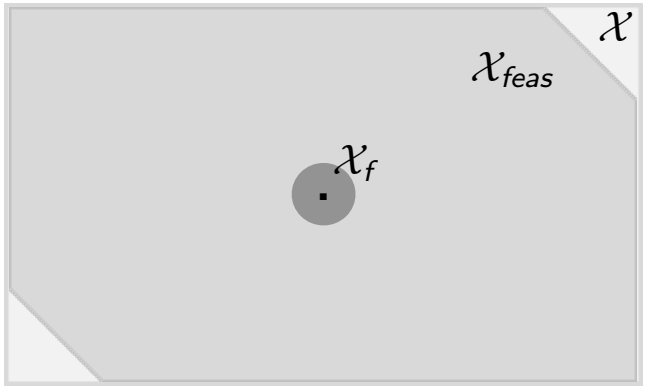
$$\approx$$

$$\pi^*(x) = \arg\min_{u \in \mathcal{U}} \ \ell(x, u) + \sum_{k=1}^{N-1} \ell(x_k, u_k) + V_f(x_N)$$

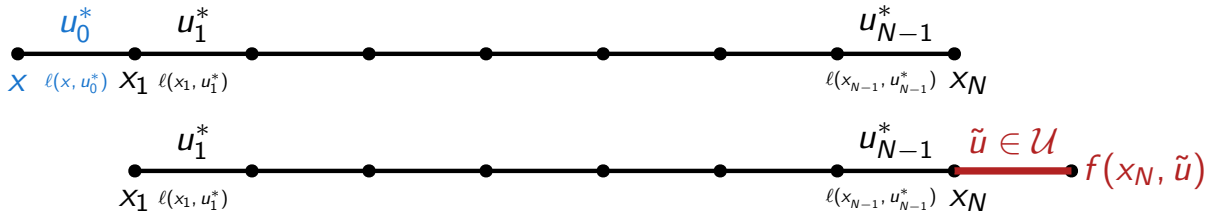
MPC - Horizon Rollout



$$
\begin{aligned}
V^*(x_t) = \min_{\mathbf{u}} \sum_{k=0}^{N-1} \ell(x_{k|t}, u_{k|t}) + V_f(x_{N|t}) \\
\text{s.t. } x_{k+1|t} = f(x_{k|t}, u_{k|t}) \\
x_{k|t} \in \mathcal{X}, u_{k|t} \in \mathcal{U} \\
x_N \in \mathcal{X}_f \\
x_{0|t} = x_t
\end{aligned}
$$



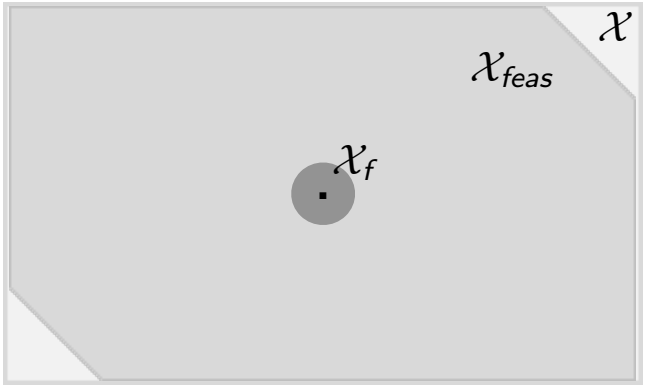
MPC - Horizon Rollout



Basic stability assumption

For all $x \in \mathcal{X}_f, \exists u \in \mathcal{U}$ satisfying

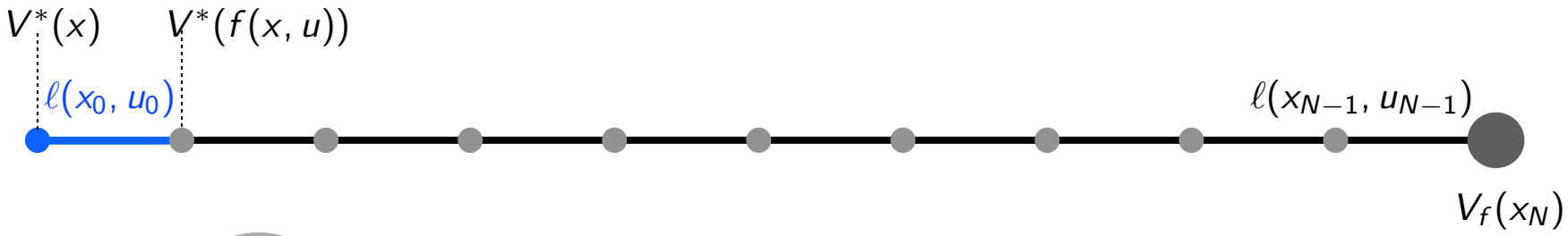
- $f(x, u) \in \mathcal{X}_f$
- $V_f(f(x, u)) - V_f(x) \leq -\ell(x, u)$,

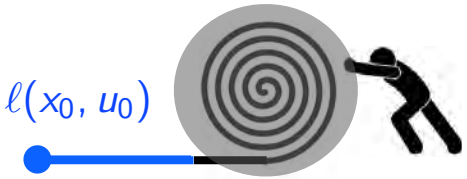


$$V_N(f(x, \pi(x))) - V_N(x) \leq -\ell(x, \pi(x)) \underbrace{-V_f(x_N) + \ell(x_N, \tilde{u}) + V_f(f(x_N, \tilde{u}))}_{\leq 0}$$

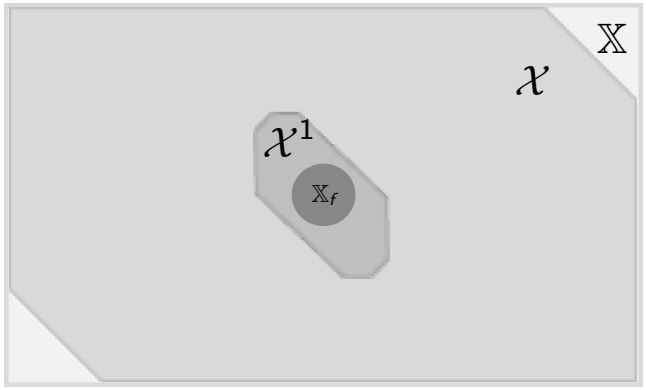
The origin is asymptotically stable under the MPC policy

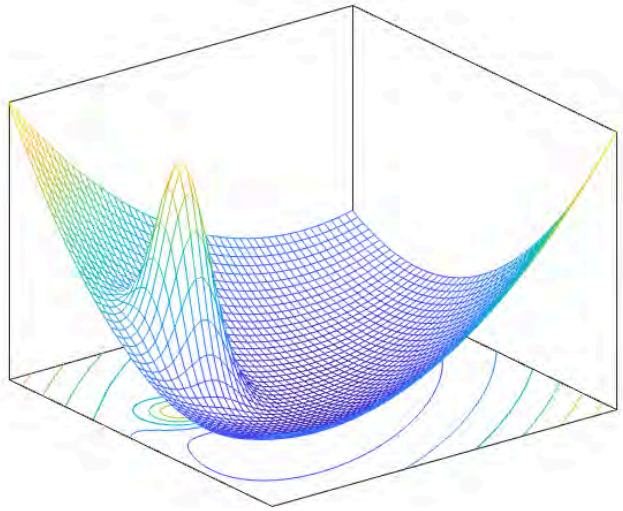
Rawlings, Mayne, Diehl, *MPC: Theory, computation, & design*, 2nd Ed., (2017)





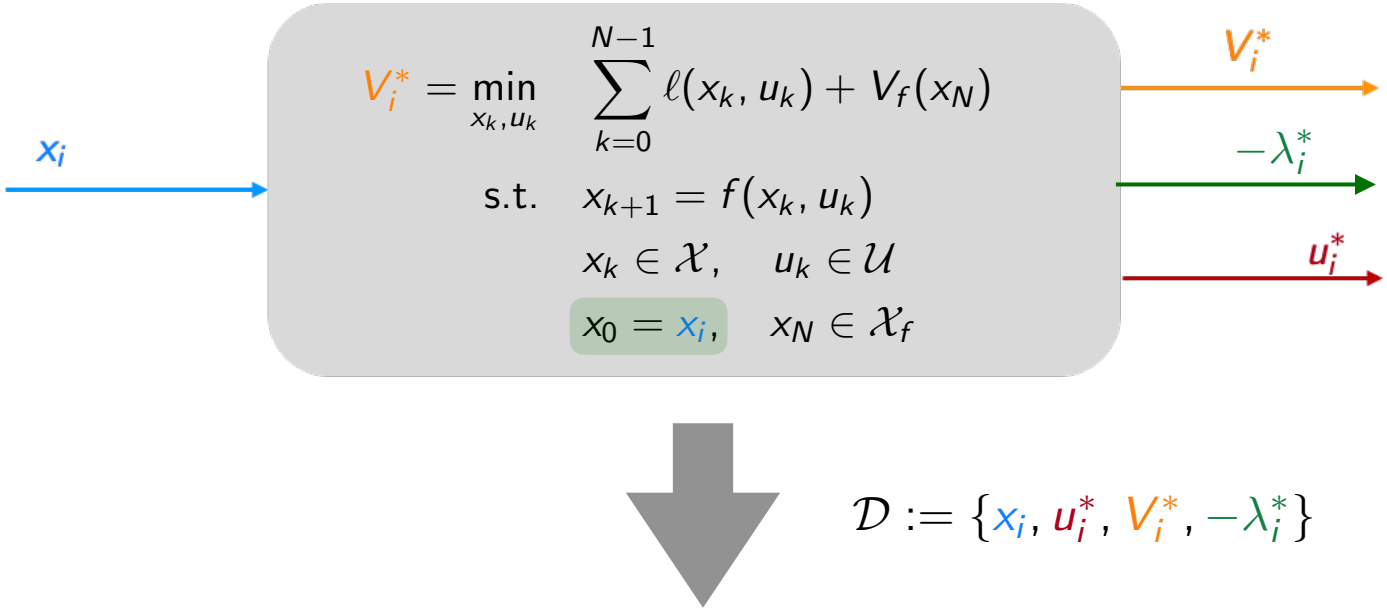
$$\min_u \ell(x_t, u) + V^*(f(x_t, u))$$
$$\text{s.t. } g(x_t, u) \leq 0$$





$$\mathcal{X}^1 = Pre(\mathbb{X}_f) := \{x \in \mathbb{X} \mid \exists u \in \mathbb{U},\ f(x,u) \in \mathbb{X}_f\}$$

Goal: Synthesize a terminal cost $\mathcal{V}_\theta(x)$



$$\mathcal{D} := \{x_i, u_i^*, V_i^*, -\lambda_i^*\}$$

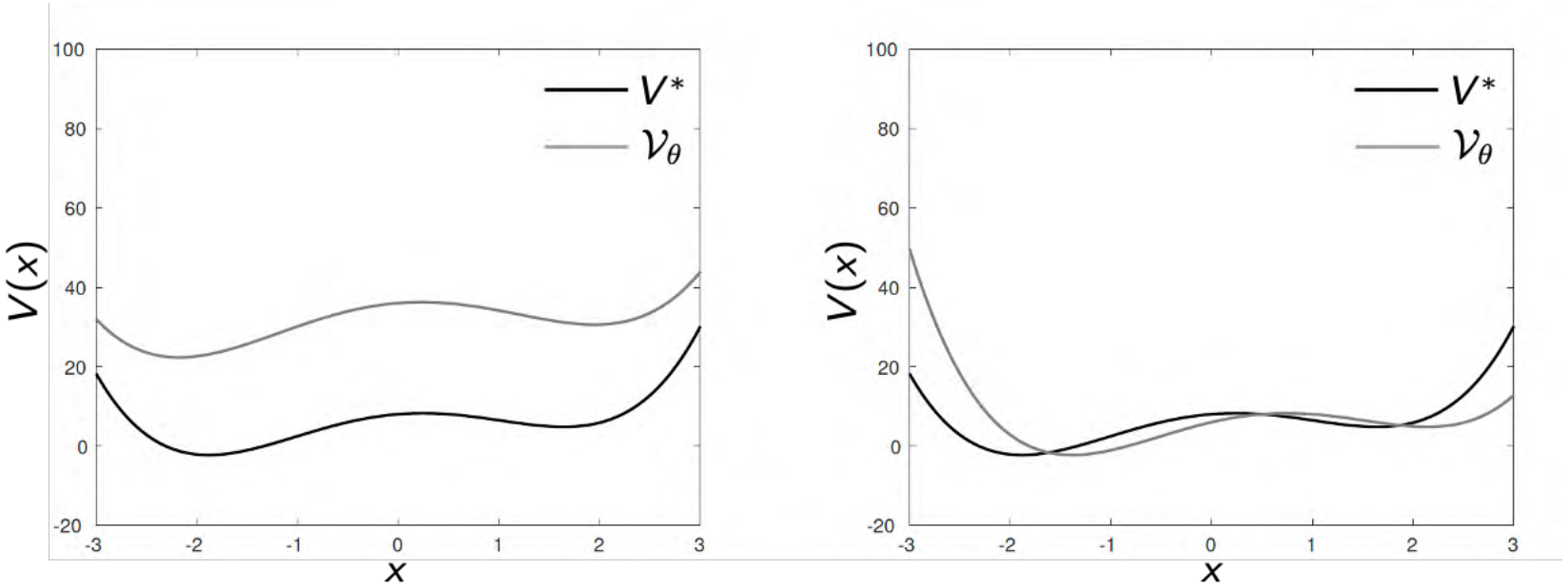
Myopic MPC

$$\begin{aligned} \min_{u \in \mathcal{U}} \quad & \ell(x, u) + \mathcal{V}_\theta(x^+) \\ \text{s.t.} \quad & x^+ = f(x, u) \in \mathcal{X} \end{aligned}$$

What constitutes a good approximation?

- $\mathcal{V}_\theta$ must be close enough to V_i^* for all i

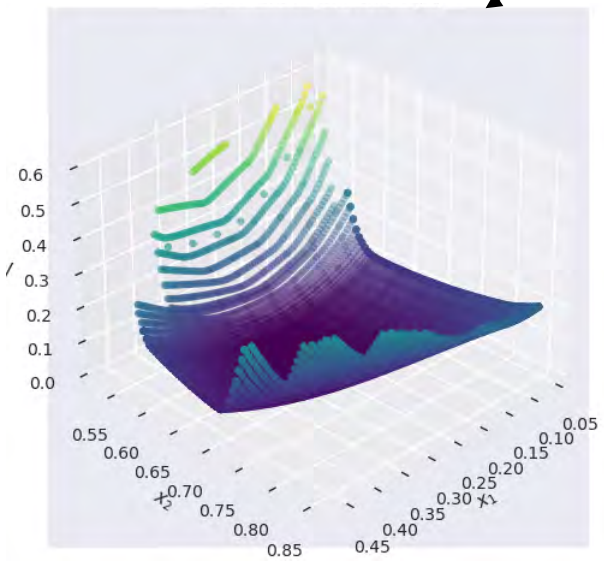
$$\hat{\theta} = \arg\min_\theta \sum_i \|\mathcal{V}_\theta(x_i) - V_i^*\|^2$$

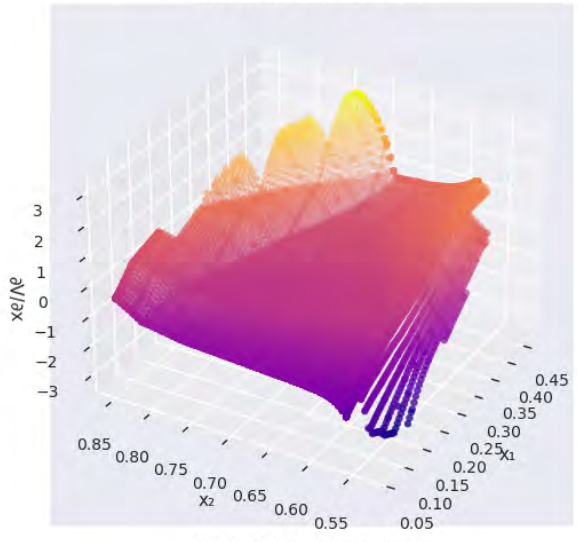


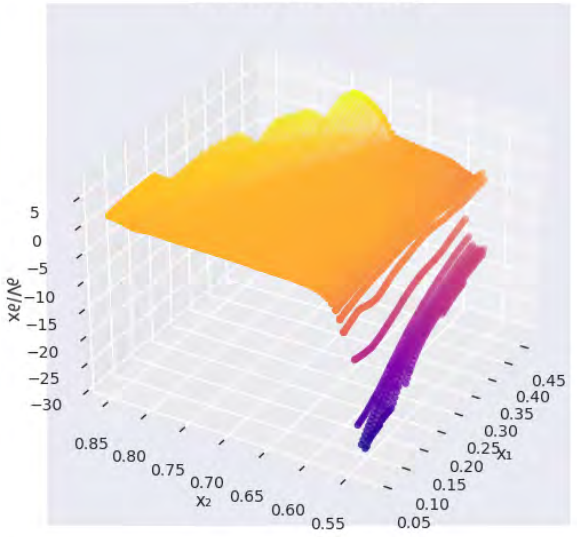
Bertsekas, *RL and optimal control* (2019)

Learning terminal costs with Sobolev training

$$\min_{\theta} \sum_{i} \left[\|\mathcal{V}_\theta(x_i) - V_i^*\|^2 + \alpha \|\nabla_x \mathcal{V}_\theta(x_i) + \lambda_i^*\|^2 \right]$$





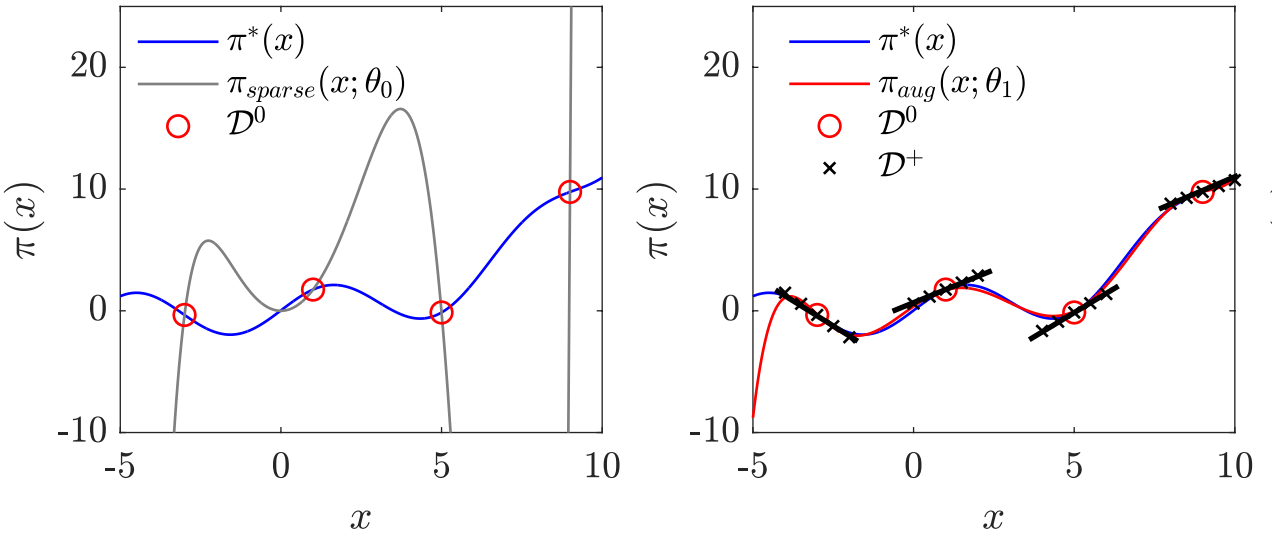


Vapnik-Chervonenkis (VC) dimension

VC-dimension measures how many data points the model can "shatter".

Shattering

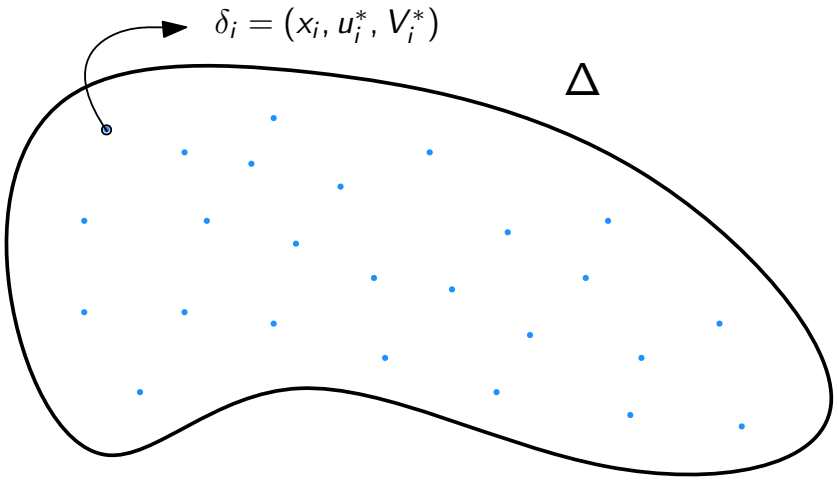
$\exists\, \theta$ for any give set of data, such that loss $\mathcal{L}(\theta) = 0$



The effective VC-dimension under Sobolev is reduced

Vidyasagar, *Learning and Generalization* (1997)

Myopic MPC: Guarantees



- Samples $\{x_i\}$ forms a ρ-net.

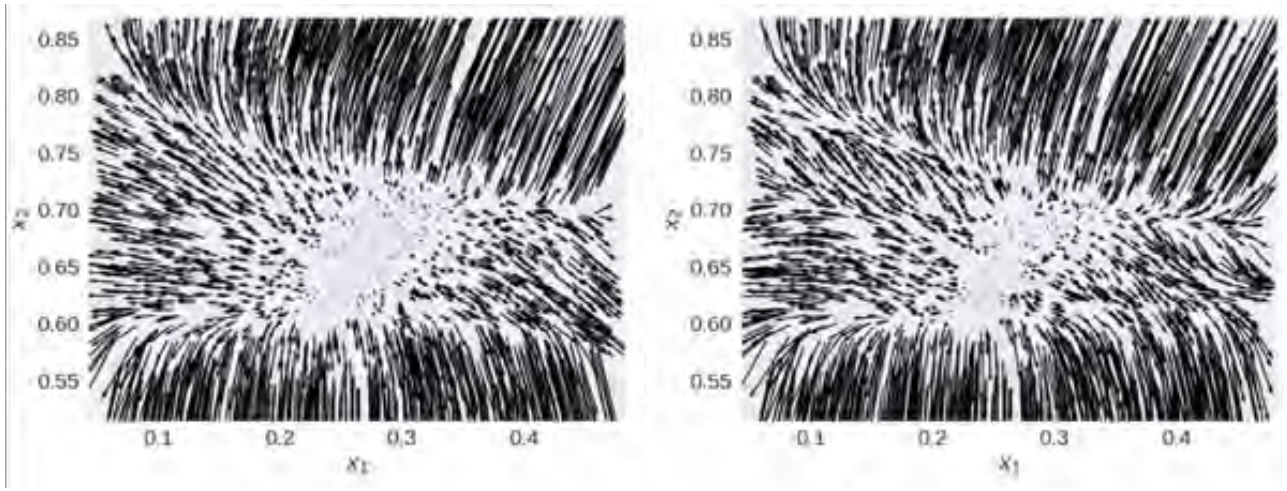
  $\exists\ i$ such that $\|x - x_i\| \leq \rho$ for all $x \in \mathcal{X}_{feas}$

- Pointwise bound at the finite samples

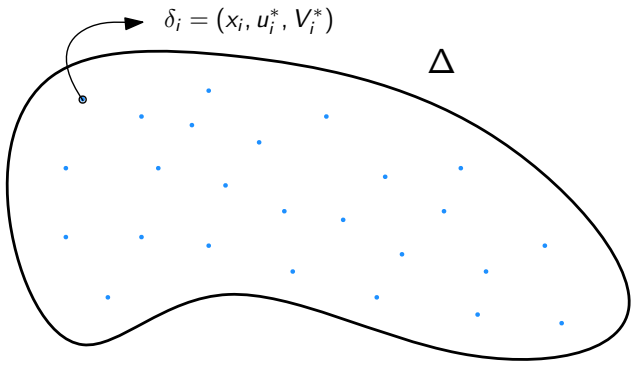
  $$\max_{1 \leq i \leq M} \|\nabla \mathcal{V}_\theta(x_i) - \nabla V^*(x_i)\| \leq d_M$$

- $\nabla \mathcal{V}(x)$ is Locally Lipschitz

$$\sup_{x \in \mathcal{X}_{feas}} \|\nabla \mathcal{V}_\theta(x) - \nabla V^*(x)\| \leq d_M + 2L\rho =: \varepsilon_g$$



Myopic MPC: Guarantees



$$\min_{\theta} \sum_{i} \left[\|\mathcal{V}_\theta(x_i) - V_i^*\|^2 + \alpha \|\nabla_x \mathcal{V}_\theta(x_i) + \lambda_i^*\|^2 \right]$$

sampling density affects ρ

gradient error term d_M

Boundedness of the policy

If LICQ and SOSC holds, the induced policy is uniformly close to the ideal policy

$$\|\pi_\theta(x) - \pi^*(x)\| \leq L_p(d_M + 2L\rho)$$

Boundedness of the closed-loop trajectories

If system dynamics are Lipschitz, the closed-loop trajectories are uniformly close

$$\|f(x, \pi_\theta(x)) - f(x, \pi^*(x))\| \leq L_f \|\pi_\theta(x) - \pi^*(x)\| \leq L_f L_p(d_M + 2L\rho)$$

Minten et al. (2026)

Myopic MPC: Guarantees

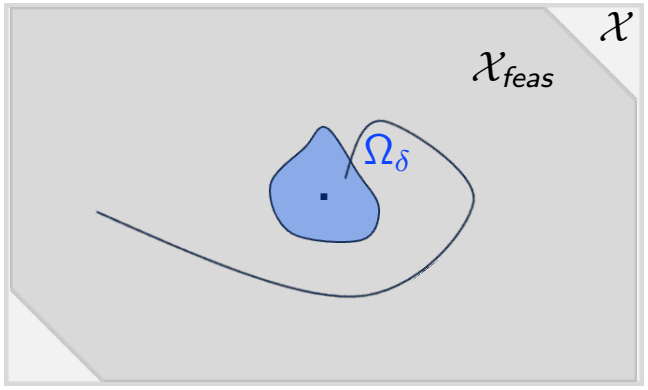
Practical stability

If $\exists$ a Lyapunov function $W : \mathcal{X}_{feas} \to \mathbb{R}_{\geq 0}$ such that $\alpha_1(\|x\|) \leq W(x) \leq \alpha_2(\|x\|)$, and $W\left(f(x, \pi^*(x))\right) - W(x) \leq -\alpha_3(\|x\|)$

$$W(f_\theta(x)) - W(x) \leq \delta - \alpha_3(\|x\|)$$

where $\delta := L_W L_f L_p(d_M + 2L\rho)$ and the trajectories converge to the residual set

$$\Omega_\delta := \{x \in \mathcal{X}_{feas} : \alpha_3(\|x\|) \leq \delta\}$$

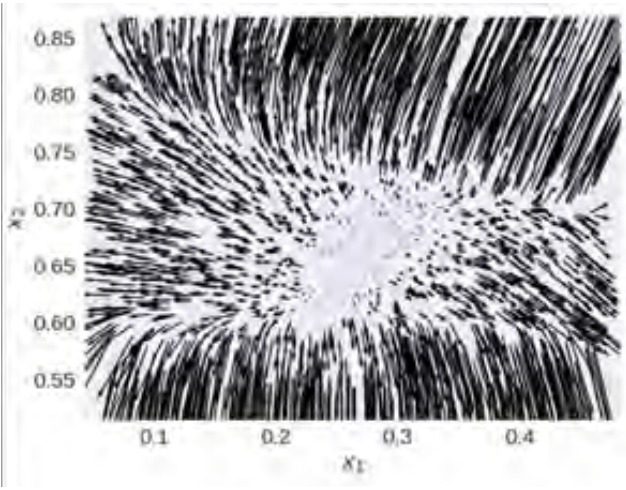


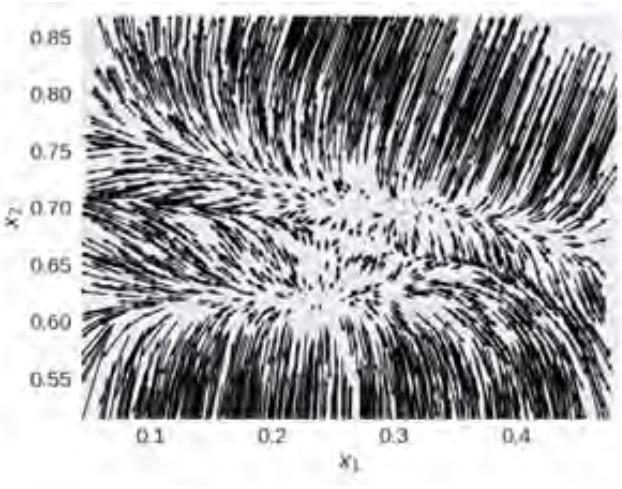
Minten et al. (2026)

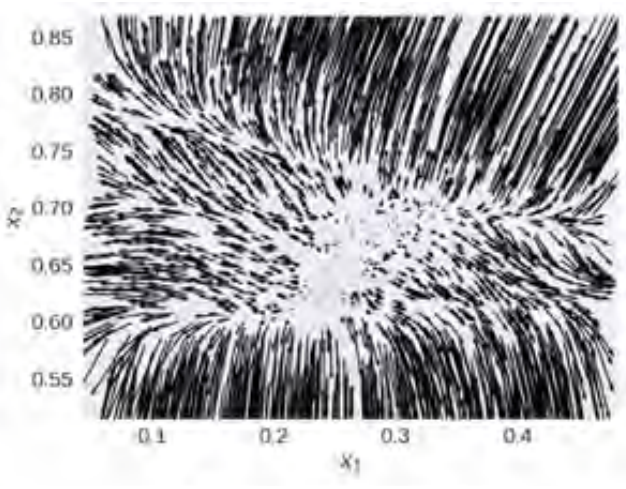
Example: Benchmark CSTR

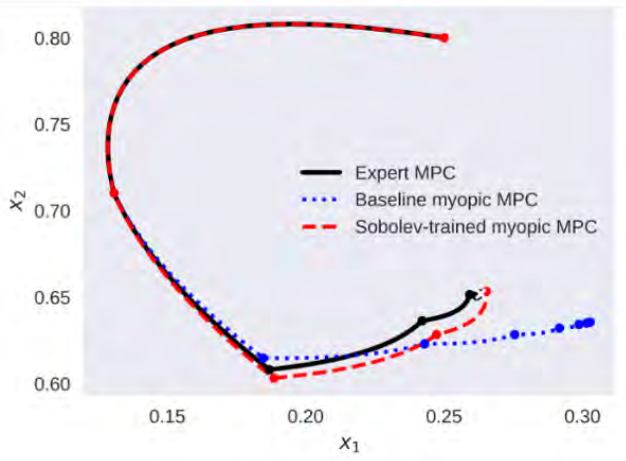
Without Sobolev ($\alpha = 0$)

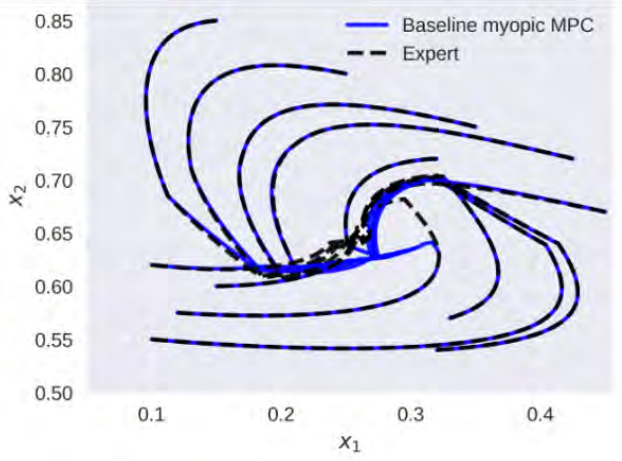
With Sobolev ($\alpha = 10$)

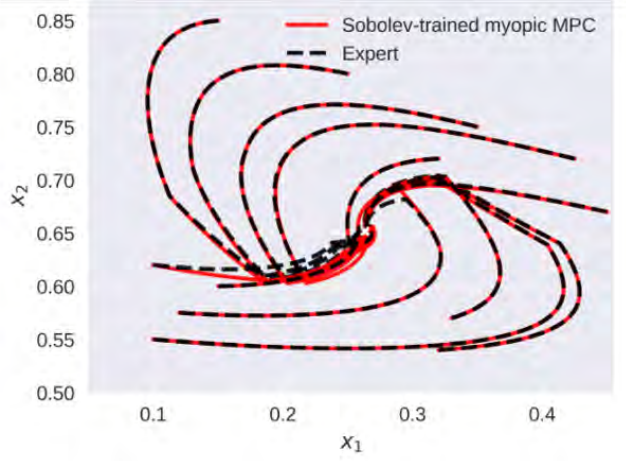












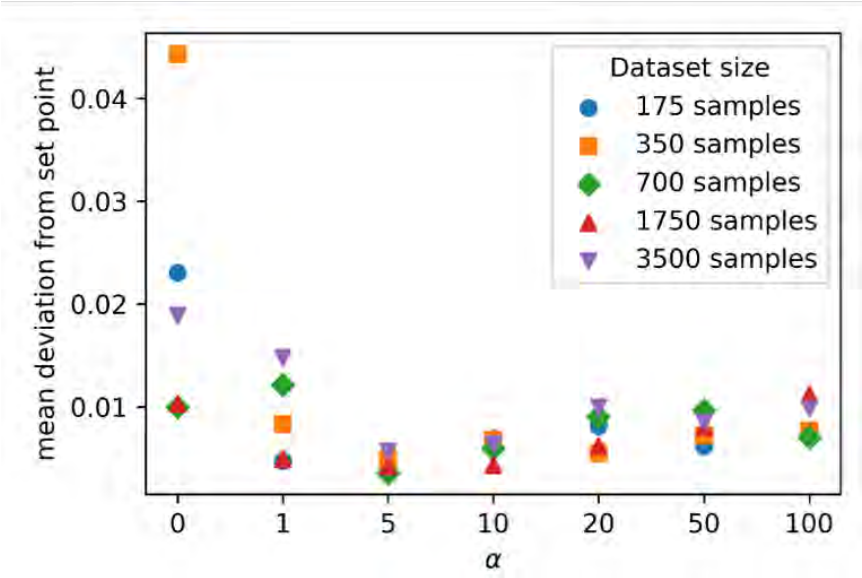
Minten et al. (2026)

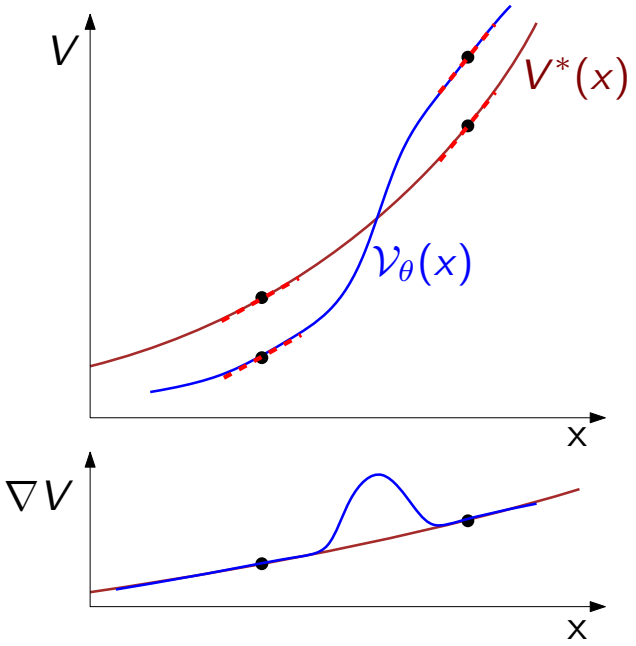
Example: Benchmark CSTR

$$\min_{\theta} \sum_{i} \left[\|\mathcal{V}_\theta(x_i) - V_i^*\|^2 + \alpha \|\nabla_x \mathcal{V}_\theta(x_i) + \lambda_i^*\|^2 \right]$$

sampling density affects ρ

gradient error term d_M



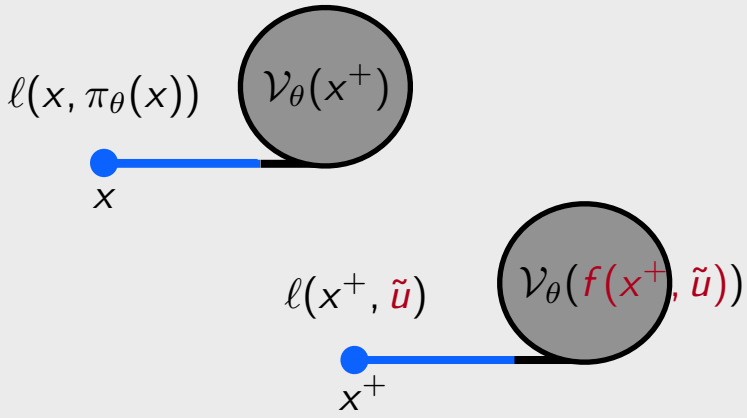


$$\delta \propto d_M + 2L\rho$$

Minten et al. (2026)

MPC nominal stability

$$\begin{aligned} \min_{u \in \mathcal{U}} \quad & \ell(x, u) + \mathcal{V}_\theta(x^+) \\ \text{s.t.} \quad & x^+ = f(x, u) \in \mathcal{X} \end{aligned}$$



$$V_1(f(x, \pi_\theta(x))) \leq V_1(x^+, \tilde{u})$$

$$V_1(f(x, \pi(x))) - V_1(x) \leq -\ell(x, \pi_\theta(x)) \underbrace{-\mathcal{V}_\theta(x^+) + \ell(x^+, \tilde{u}) + \mathcal{V}_\theta(f(x^+, \tilde{u}))}_{\leq 0 \ \forall x \in \mathcal{X}_{feas}?}$$

What constitutes a good approximation?

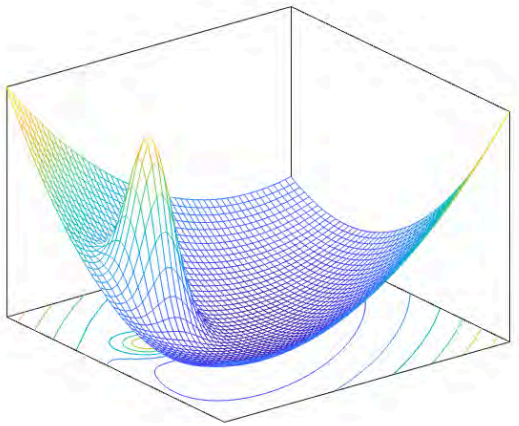
Desirable property

- Upon applying optimal u_i^* at x_i,
- cost-to-go from the successor state $\mathcal{V}_\theta(f(x_i, u_i^*))$ must be lower than $\mathcal{V}_\theta(x_i)$

Learn with descent property constraint

$$\begin{aligned} \hat{\theta} = \arg\min_{\theta} \quad & \sum_i \|\mathcal{V}_\theta(x_i) - V_i^*\|^2 \\ \text{s.t.} \quad & \mathcal{V}_\theta(f(x_i, u_i^*)) - \mathcal{V}_\theta(x_i) \leq -\ell(x_i, u_i^*), \quad \forall i \end{aligned}$$

- Special case: $\mathcal{V}_\theta(x) := x^\top P x$, with $P = L^\top L$



Abdufattokhov et al. *IEEE CDC* (2021);

Learn terminal cost with descent property constraint

Semi-infinite synthesis problem

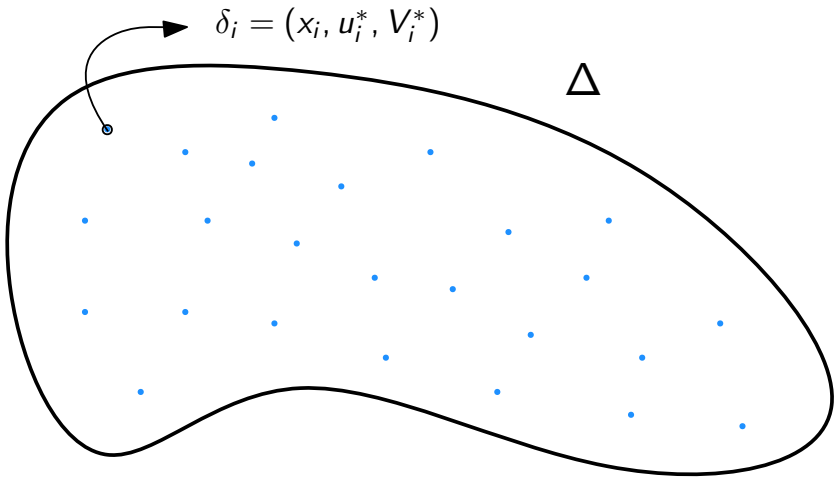
$$\min_{\theta \in \mathbb{R}^d} \quad \sum_i \mathcal{G}\left(\mathcal{V}_\theta(x), V\right)$$

$$\text{s.t.} \quad \mathcal{V}_\theta\left(f(x,u)\right) - \mathcal{V}_\theta(x) \leq -\ell(x,u), \quad \forall x \in \mathcal{X}_{feas}$$

Scenario problem

$$SP_M : \min_{\theta \in \mathbb{R}^d} \sum_i \mathcal{G}\left(\mathcal{V}_\theta(x_i), V_i^*\right)$$

$$\text{s.t.} \quad \mathcal{V}_\theta\left(f(x_i, u_i^*)\right) - \mathcal{V}_\theta(x_i) \leq -\ell(x_i, u_i^*), \quad \forall i = 1, \ldots, M$$

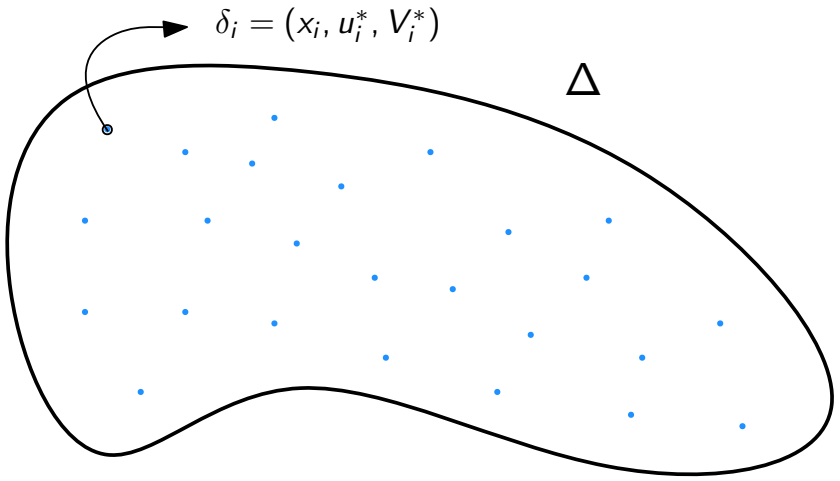


- For all $\mathcal{D} = \{\delta_i\} \in \Delta$, we have $\theta \in \Theta_\delta$
- What can we say about constraint satisfaction for any $\delta \in \Delta$?

Learn terminal cost with descent property constraint

Scenario problem

$$SP_M : \min_{\theta \in \mathbb{R}^d} c^\top \theta,$$
$$\text{s.t. } \theta \in \Theta_{\delta^{(i)}},$$
$$\forall \delta^{(i)}, i = 1, \dots, M,$$



- For all $\mathcal{D} = \{\delta_i\} \in \Delta$, we have $\theta \in \Theta_\delta$
- What can we say about constraint satisfaction for any $\delta \in \Delta$?

Learn terminal cost with descent property constraint

$$SP_M : \min_{\theta \in \mathbb{R}^d} c^\top \theta,$$
$$\text{s.t. } \theta \in \Theta_{\delta^{(i)}},$$
$$\forall \delta^{(i)}, i = 1, \dots, M,$$

- SP_M is convex in θ
- $\{\delta_i\}$ are i.i.d samples from Δ
- SP_M is feasible and solution θ^* exists and is unique

Violation probability

For a given $\theta \in \Theta$, violation probability is defined as

$$\mathcal{P}(\theta) = \mathbb{P}\{\delta \in \Delta : \theta \notin \Theta_\delta\}$$

Generalization theorem of the Scenario Problem

Under these assumptions the following holds

$$\mathbb{P}^M \left\{\mathcal{P}\left(\theta_M^*\right) > \epsilon\right\} \leq \sum_{i=0}^{d-1} \binom{M}{i} \epsilon^i (1-\epsilon)^{M-i}$$

Campi & Garatti, *SIAM J. on Opt.* (2008)

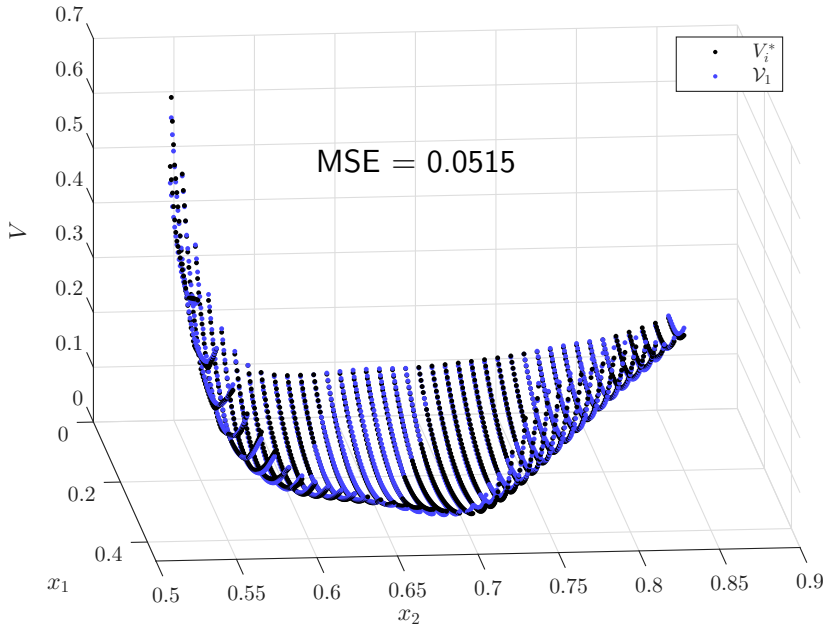
Probabilisitic certificate for the violation of the descent property

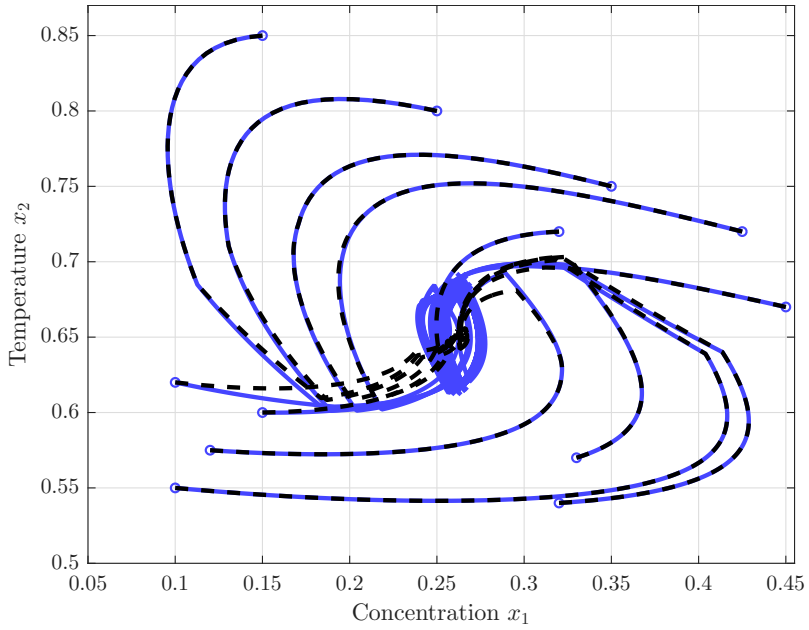
$$\mathbb{P}\{\mathcal{V}(f(x, \kappa_N(x)); \theta_M^*) - \mathcal{V}(x; \theta_M^*) \leq -\ell(x, \kappa_N(x))\} > \epsilon$$

Baltussen et al. *IEEE CDC* (2025)

Example: Benchmark CSTR

$$\min_{\theta} \quad \sum_{i} \|\mathcal{V}_\theta(x_i) - V_i^*\|^2$$

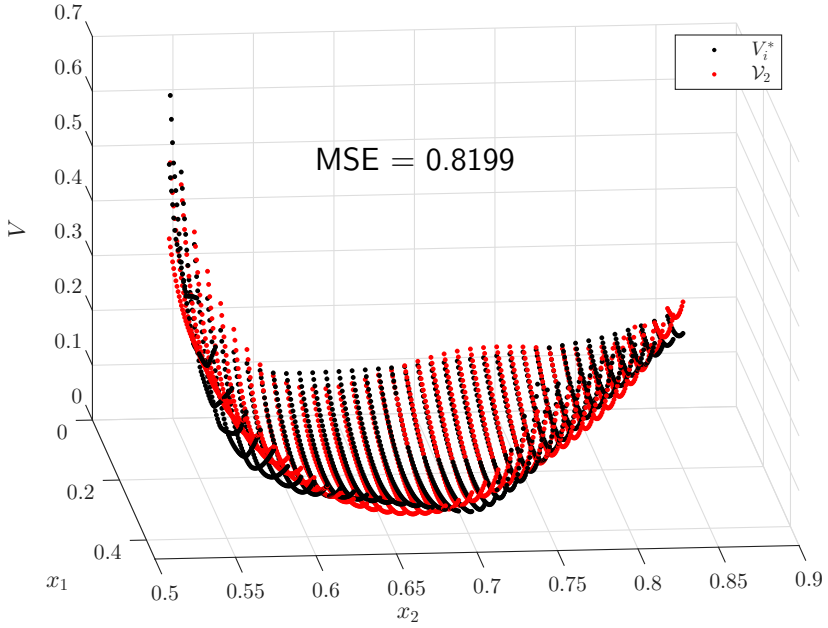


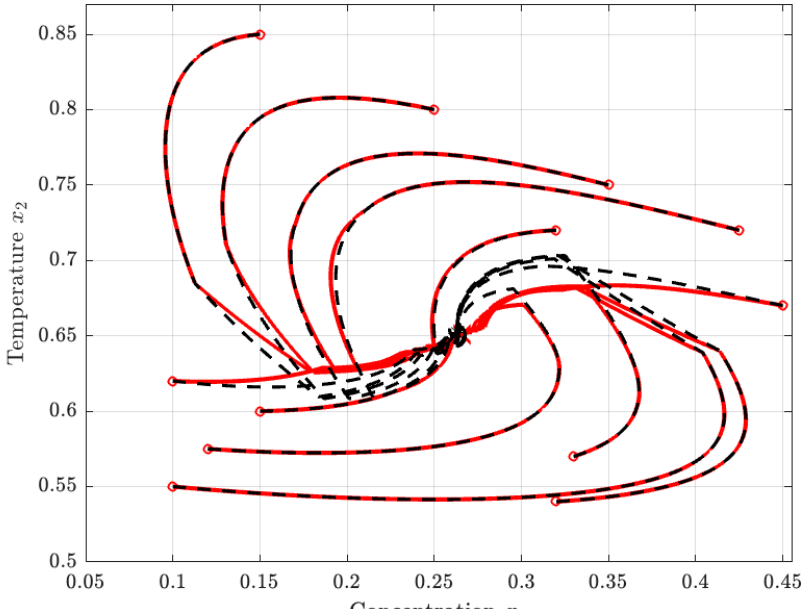


Example: Benchmark CSTR

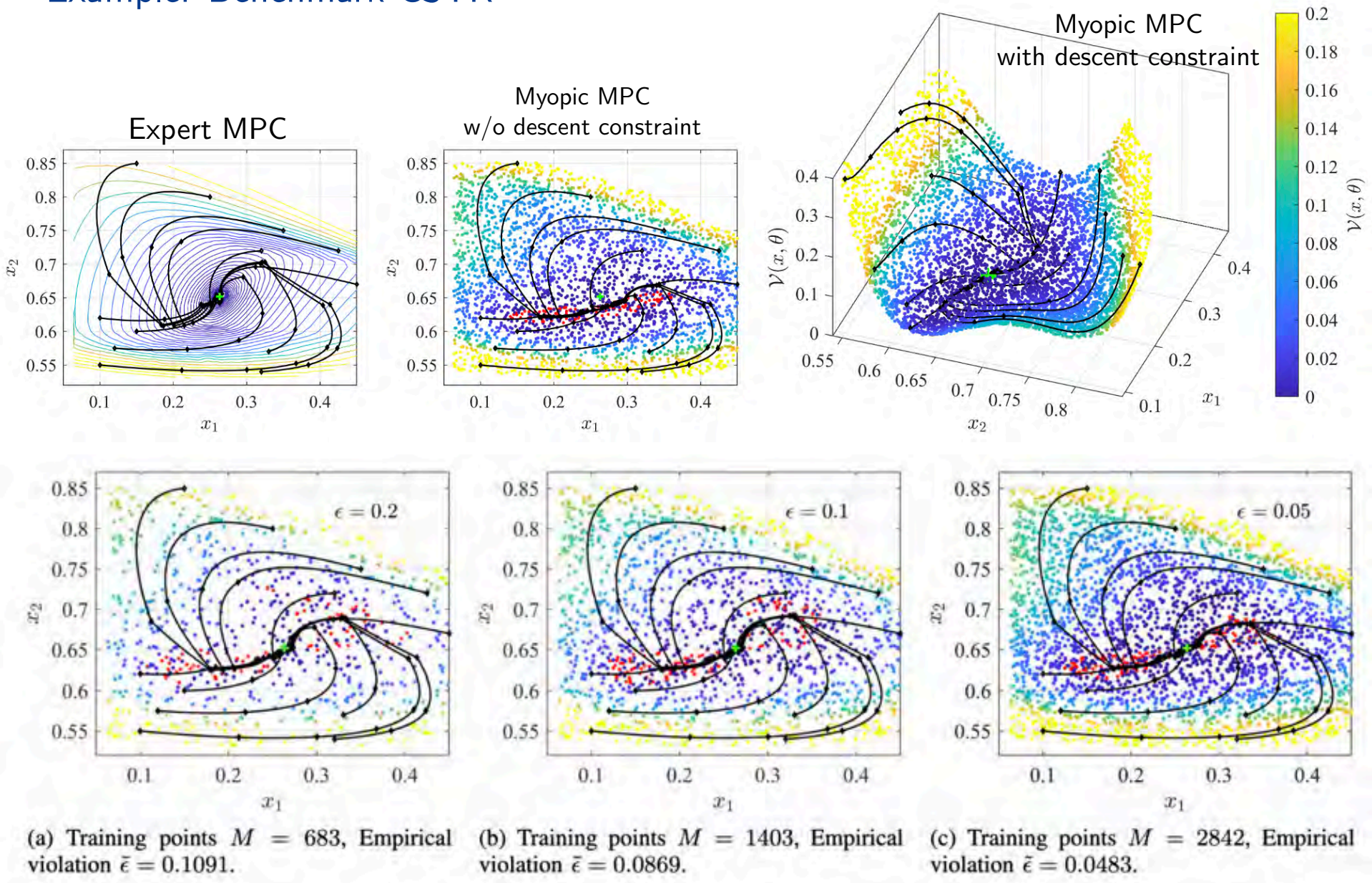
$$\min_{\theta} \quad \sum_i \|\mathcal{V}_\theta(x_i) - V_i^*\|^2$$

$$\text{s.t.} \quad \mathcal{V}_\theta(f(x_i, u_i^*)) - \mathcal{V}_\theta(x_i) \leq -\ell(x_i, u_i^*)$$





Example: Benchmark CSTR



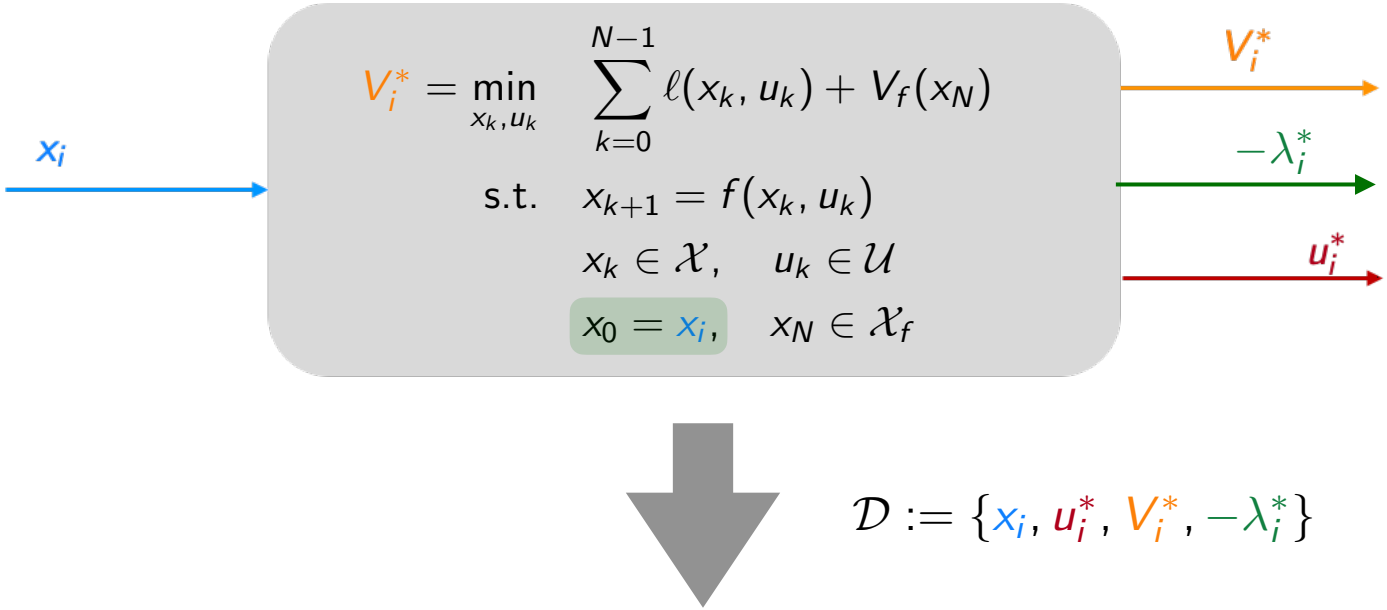
(a) Training points $M = 683$, Empirical violation $\bar{\epsilon} = 0.1091$.

(b) Training points $M = 1403$, Empirical violation $\bar{\epsilon} = 0.0869$.

(c) Training points $M = 2842$, Empirical violation $\bar{\epsilon} = 0.0483$.

Baltussen et al. *IEEE CDC* (2025)

Goal: Synthesize a terminal cost $\mathcal{V}_\theta(x)$

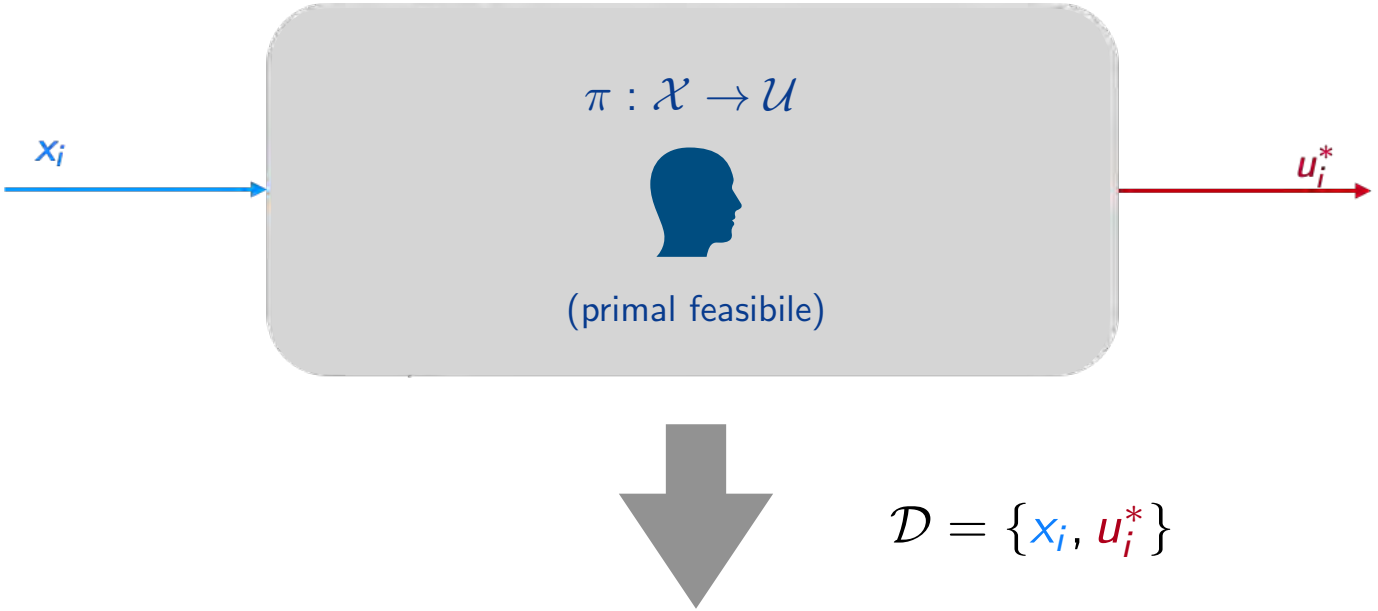


$$\mathcal{D} := \{x_i, u_i^*, V_i^*, -\lambda_i^*\}$$

Myopic MPC

$$\begin{aligned} \min_{u \in \mathcal{U}} \quad & \ell(x, u) + \mathcal{V}_\theta(x^+) \\ \text{s.t.} \quad & x^+ = f(x, u) \in \mathcal{X} \end{aligned}$$

Goal: Synthesize terminal cost from $\mathcal{D} = \{x_i, u_i^*\}$



Myopic MPC

$$\begin{aligned} \min_{u \in \mathcal{U}} \quad & \ell(x, u) + \mathcal{V}_\theta(x^+) \\ \text{s.t.} \quad & x^+ = f(x, u) \in \mathcal{X} \end{aligned}$$

Learning terminal cost from expert demonstrations $\mathcal{D} = \{x_i, u_i^*\}$

Myopic MPC

$$\begin{aligned} \min_{u} \quad & \ell(x, u) + \mathcal{V}_\theta(f(x, u)) \\ \text{s.t.} \quad & g(x, u) \leq 0 \end{aligned}$$

$$\mathcal{L}(x, u, \lambda) := \ell(x, u) + \mathcal{V}_\theta(f(x, u)) + \lambda^\top g(x, u) \qquad \text{(Lagrangian)}$$

KKT conditions of the myopic MPC

$$\begin{aligned} \nabla_u \ell(x, u) + \nabla_u f(x, u)^\top \nabla_x \mathcal{V}_\theta(f(x, u)) + \nabla_u g(x, u)^\top \lambda = 0 \quad & \text{(stationarity)} \\ \lambda \circ g(x, u) = 0 \quad & \text{(complementary slackness)} \\ \lambda \geq 0 \quad & \text{(Dual feasibility)} \end{aligned}$$

For what values of θ, do the data points $\mathcal{D} = \{(x_i, u_i^*)\}$,
satisfy the KKT conditions of the myopic MPC?

Learning terminal cost from expert demonstrations $\mathcal{D} = \{x_i, u_i^*\}$

For what values of θ, do the data points $\mathcal{D} = \{(x_i, u_i^*)\}$,
satisfy the KKT conditions of the myopic MPC?

KKT residuals

$$r_{stat}^{(i)}(\theta, \lambda_i) := \nabla_u \ell(x_i, u_i^*) + \nabla_u f(x_i, u_i^*)^\top \nabla_x \mathcal{V}_\theta(f(x_i, u_i^*)) + \nabla_u g(x_i, u_i^*)^\top \lambda_i$$
$$r_{comp}^{(i)}(\theta, \lambda_i) := \lambda_i \circ g(x_i, u_i^*)$$

KKT residual minimization

$$\min_{\theta, \{\lambda_i\}} \quad \sum_i \|r_{stat}^{(i)}(\theta, \lambda_i)\|^2 + \|r_{comp}^{(i)}(\theta, \lambda_i)\|^2 \qquad \text{(Stationarity \& complementary slackness)}$$
$$\text{s.t.} \quad \lambda_i \geq 0, \quad \forall i \qquad \text{(Dual feasibility)}$$
$$\mathcal{V}_\theta(f(x_i, u_i^*)) - \mathcal{V}_\theta(x_i) \leq -\ell(x_i, u_i^*), \quad \forall i \qquad \text{(descent property constraint)}$$

Keshavraz, Wang, Boyd, *IEEE MCSC* (2011)

KKT residual minimization for mixed-integer MPC

- Formulate myopic MPC with integer decisions
- Relax integer requirements
- Evaluate KKT residuals of the relaxed problem at finite samples $\mathcal{D} = \{x_i, u_i^*\}$
- Synthesize terminal cost $\mathcal{V}_\theta(x)$ by minimizing KKT residuals
- Use this in the myopic MPC with integer decisions

Myopic MPC

$$\begin{aligned} \min_{u} \quad & \ell(x, u) + \mathcal{V}_\theta(f(x, u)) \\ \text{s.t.} \quad & g(x, u) \leq 0 \end{aligned}$$

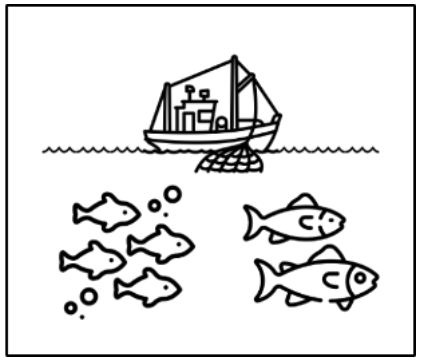
KKT residual minimization

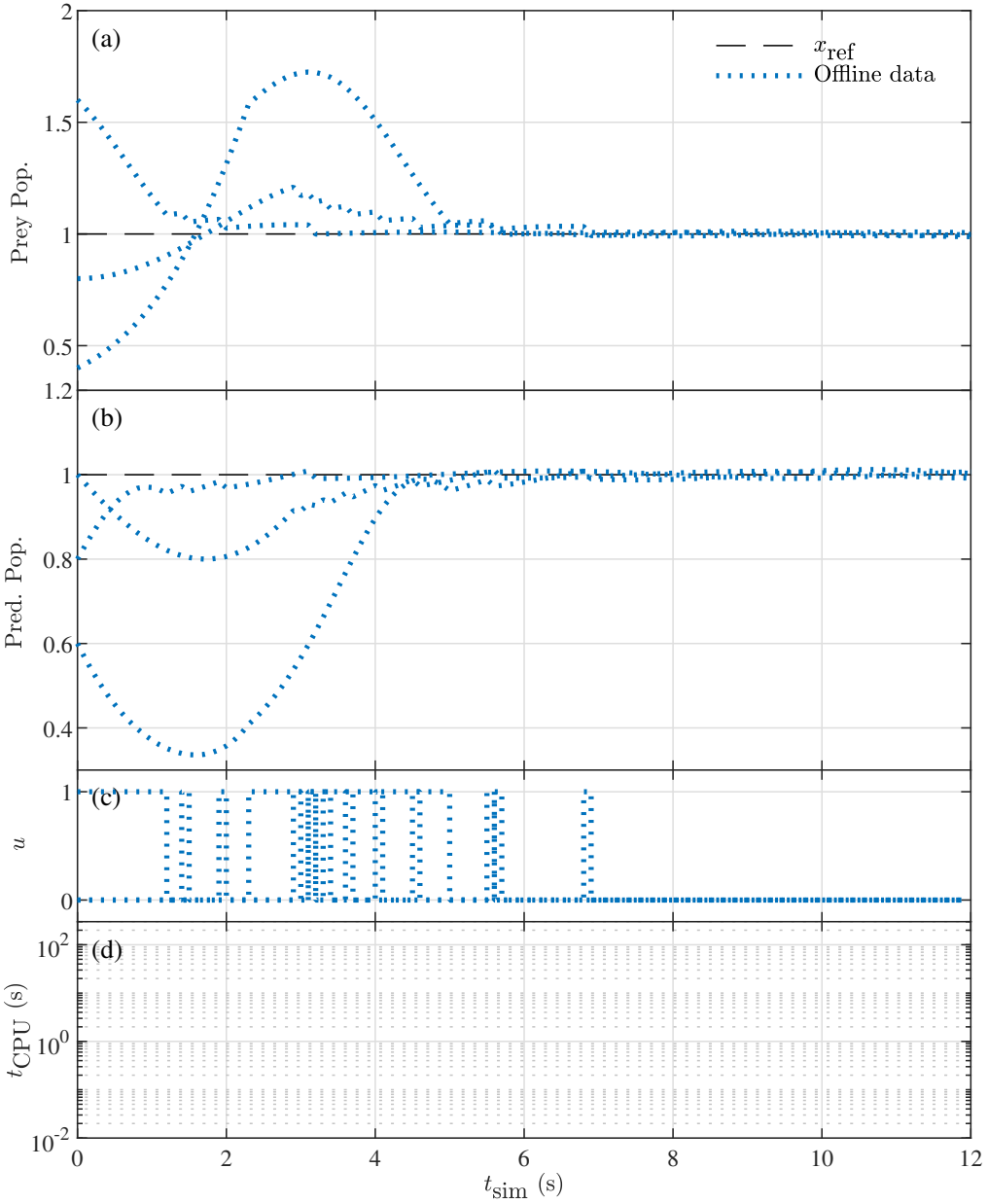
$$\begin{aligned} \min_{\theta, \{\lambda_i\}} \quad & \sum_i \|r_{stat}^{(i)}(\theta, \lambda_i)\|^2 + \|r_{comp}^{(i)}(\theta, \lambda_i)\|^2 && \text{(Stationarity \& complementary slackness)} \\ \text{s.t.} \quad & \lambda_i \geq 0, \quad \forall i && \text{(Dual feasibility)} \\ & \mathcal{V}_\theta(f(x_i, u_i^*)) - \mathcal{V}_\theta(x_i) \leq -\ell(x_i, u_i^*), \quad \forall i && \text{(descent property constraint)} \end{aligned}$$

Orrico et al. (2026)

Mixed-integer Myopic MPC using KKT residual minimzation

Lotka-Volterra problem

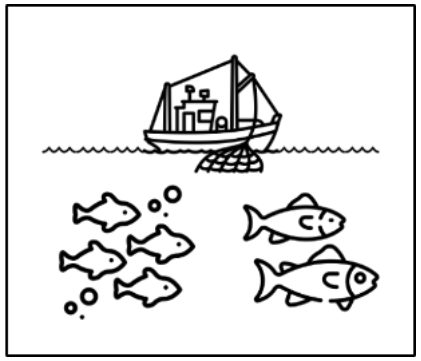


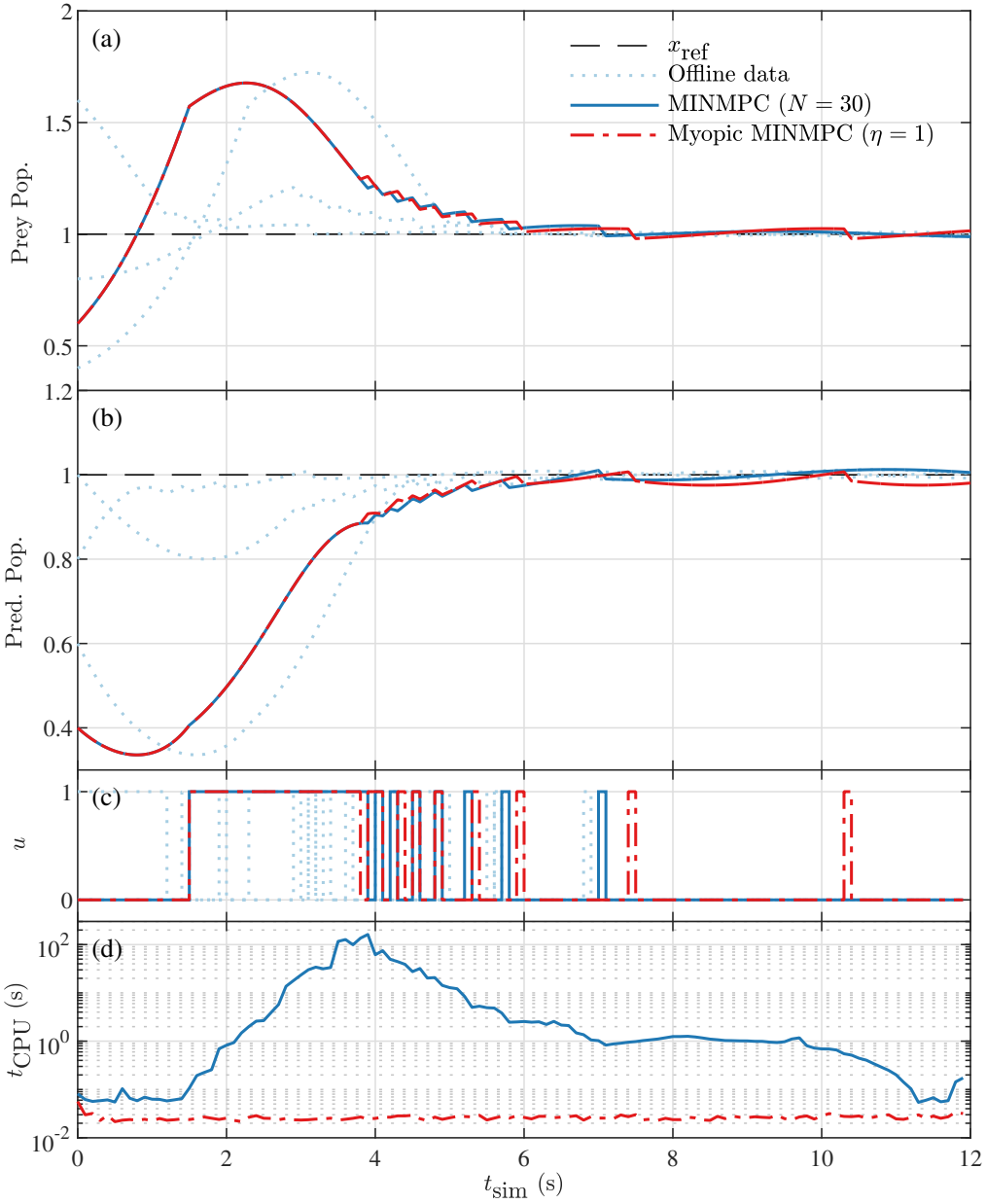


Orrico et al. (2026)

Mixed-integer Myopic MPC using KKT residual minimzation

Lotka-Volterra problem

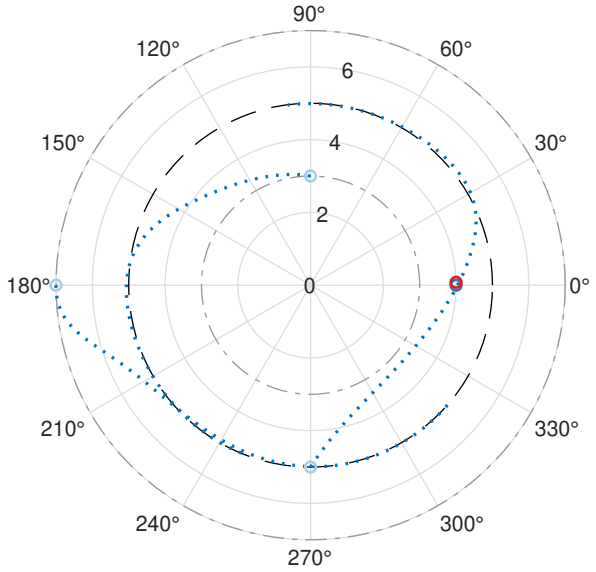


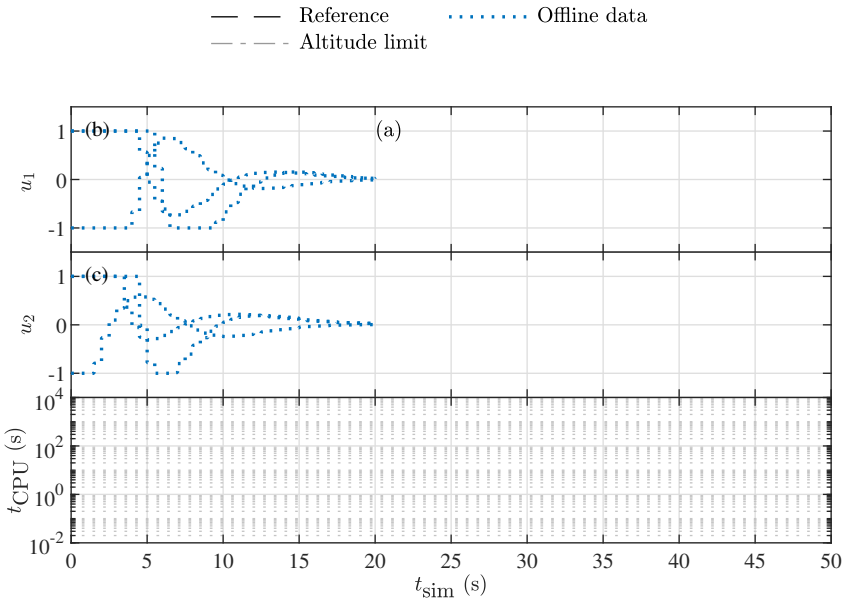


Orrico et al. (2026)

Mixed-integer Myopic MPC using KKT residual minimzation

Satellite in orbit

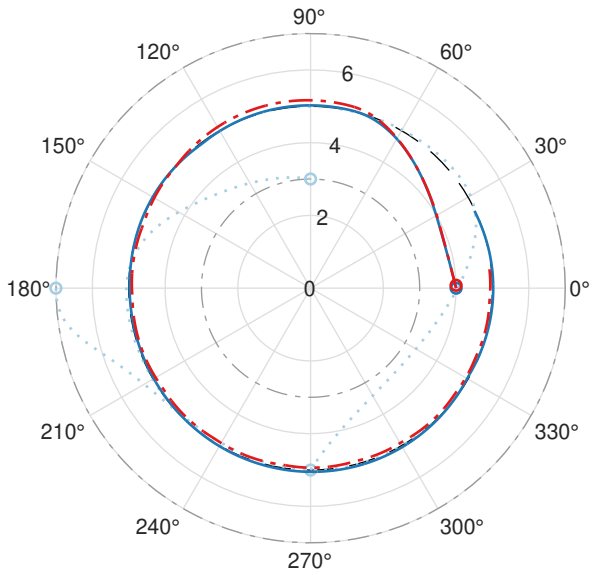


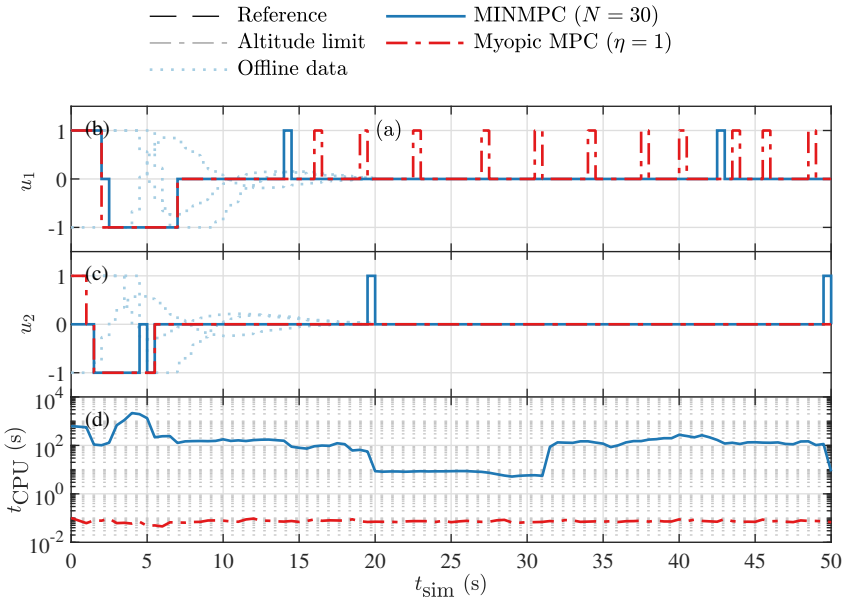


Orrico et al. (2026)

Mixed-integer Myopic MPC using KKT residual minimzation

Satellite in orbit

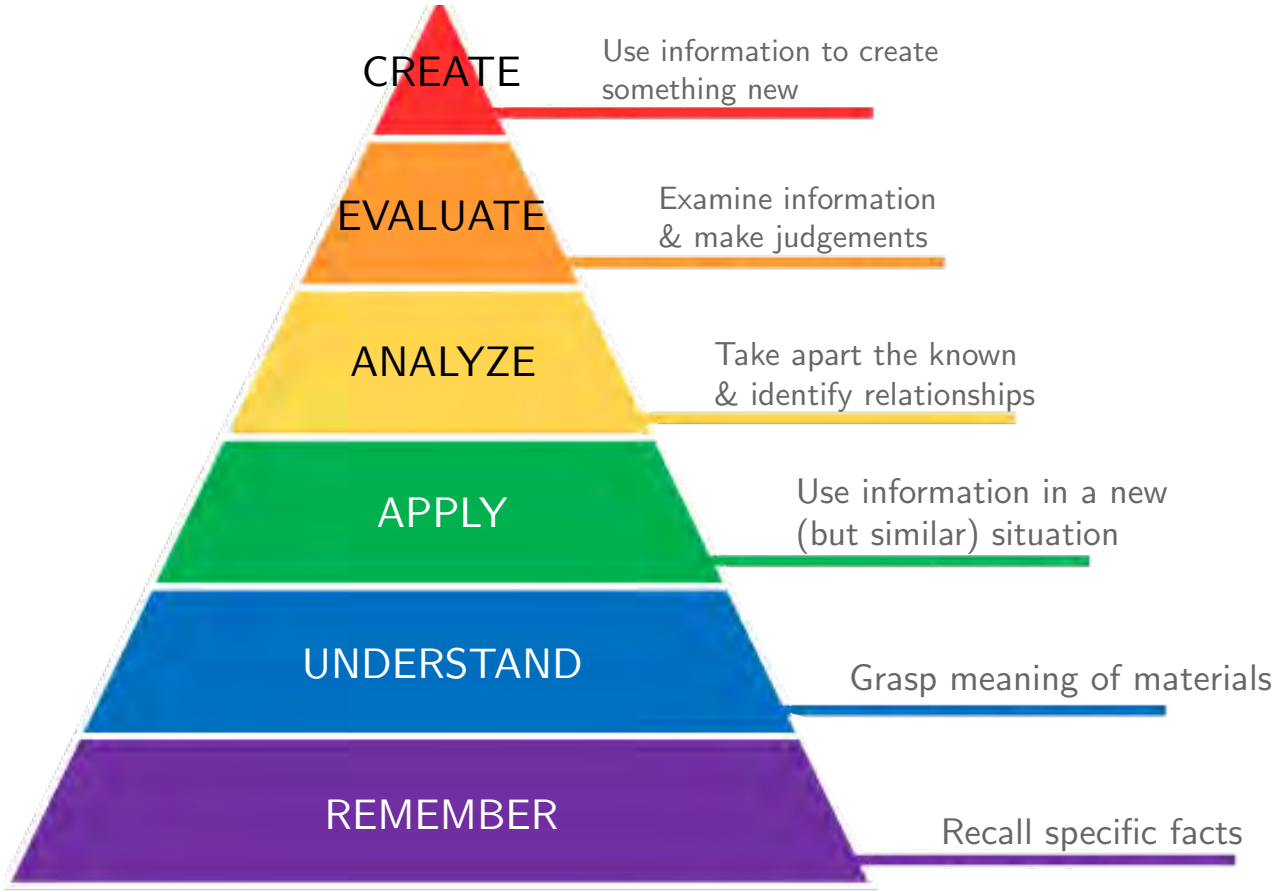




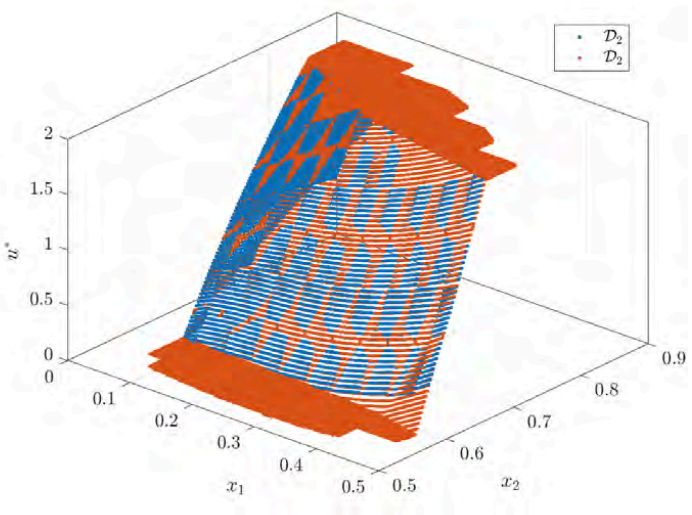
Orrico et al. (2026)

Bloom's Cognitive Taxonomy

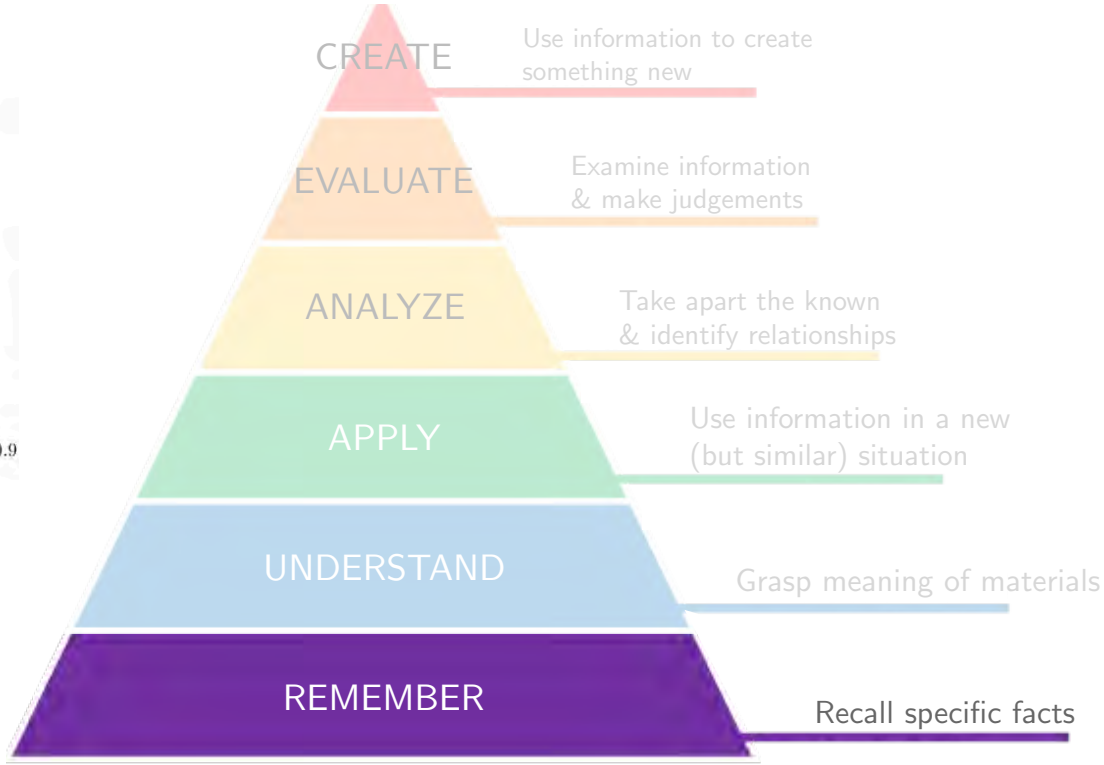
- Learning is a process by which an agent's intelligence develops through experience or data.
- Cognition is the mechanism through which intelligence manifests



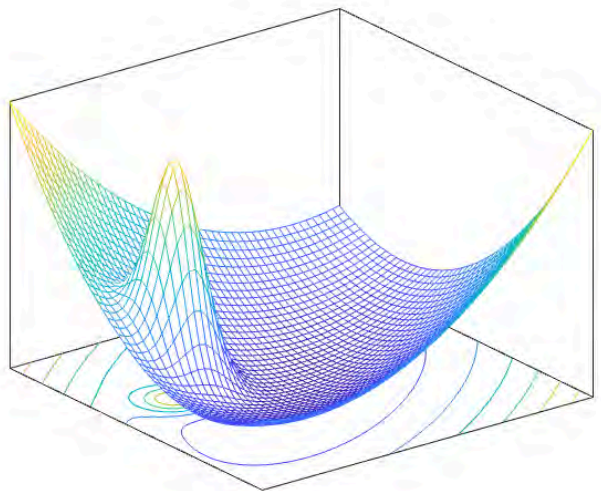
Policy approximation



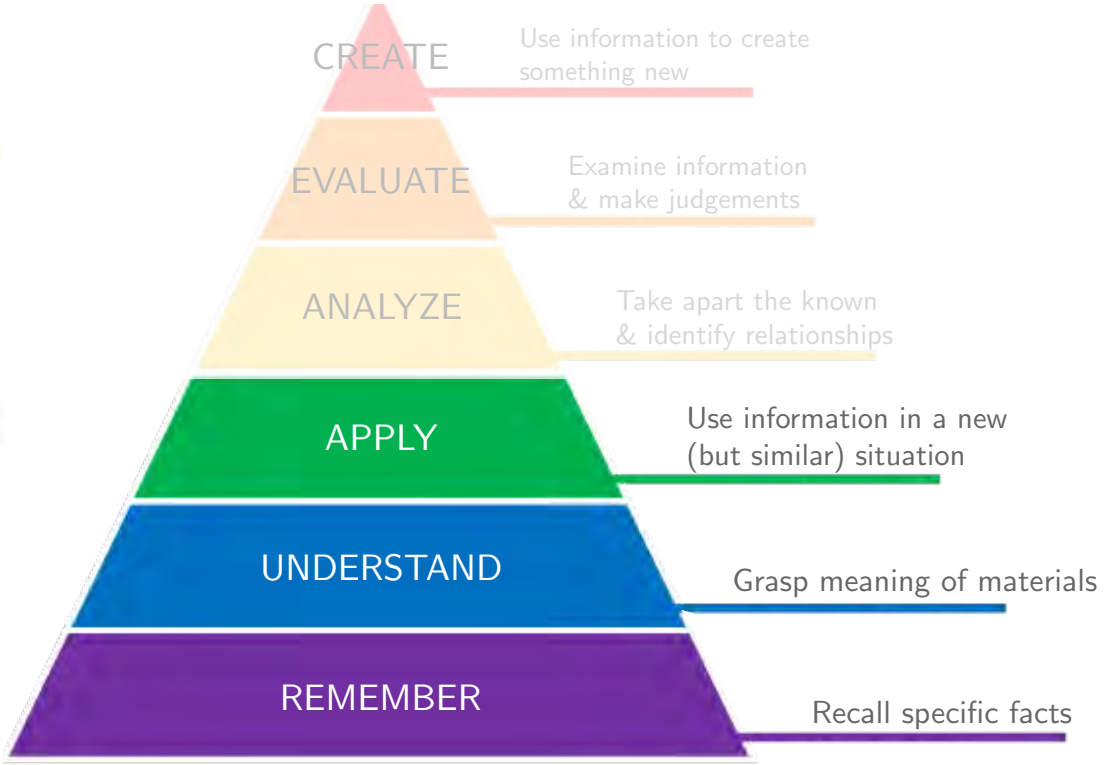
$$\pi_\theta : \mathbb{R}^n \to \mathbb{R}^m$$



Value function approximation

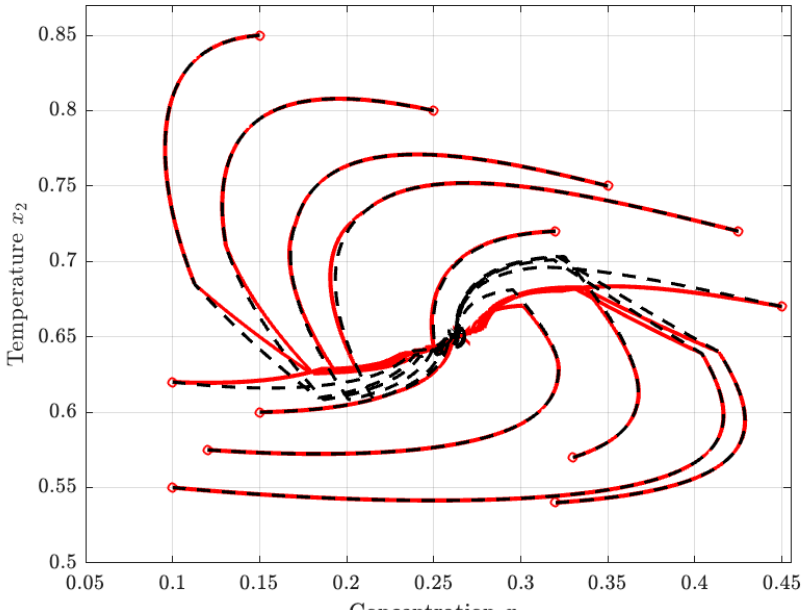


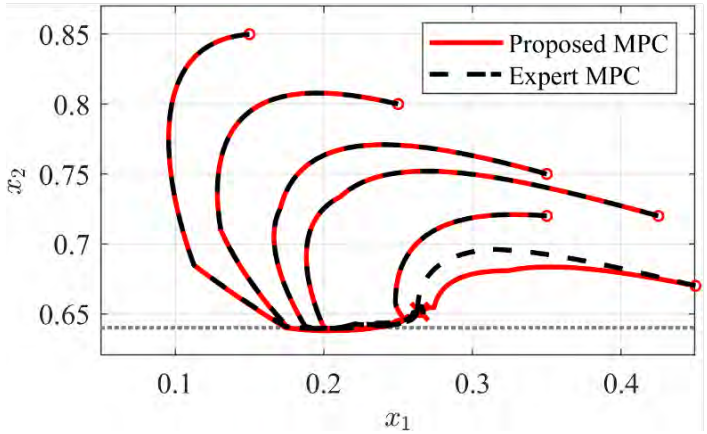
$$\begin{aligned} \min_{u} \quad & \ell(x,u) + \mathcal{V}_\theta(f(x,u)) \\ \text{s.t.} \quad & g(x,u) \leq 0 \end{aligned}$$



APPLY

Use information in a new (but similar) situation





- Tightened constraints
- Sampling time: $3 \rightarrow 0.1$s

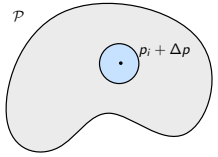
Value function approximation facilitates flexibility

Concluding Remarks

	Reinforcement Learning	Supervised Learning
Approximate policy	Policy gradient	DNN Explicit MPC
Approximate Value function	Approx. DP/ Q-learning	Myopic MPC

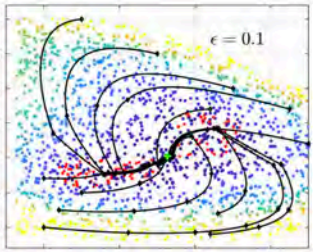
- Requires high fidelity simulators or real system
- Several episodes to learn

- Accurate representation of policy crucial
- No changes to controller parameters



$$\min_{\theta} \sum_i \left[\|\mathcal{V}_\theta(x_i) - V_i^*\|^2 + \alpha \|\nabla_x \mathcal{V}_\theta(x_i) + \lambda_i^*\|^2 \right]$$

$$\text{s.t.} \quad \mathcal{V}_\theta(f(x_i, u_i^*)) - \mathcal{V}_\theta(x_i) \leq -\ell(x_i, u_i^*), \quad \forall i$$





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